Reference Dependence and the Role of Information Frictions[†]

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Abstract

We introduce positive and negative endowment shocks and information frictions into an experimental labor market and study how firms share these with workers. Workers evaluate wages consistently with a reference wage, demanding higher (lower) wages but exerting the same effort when a positive (negative) shock is common knowledge. Their wage expectations adjust instantaneously to new information but sluggishly and asymmetrically when unaware of the shock. Firms form accurate beliefs about how shocks and information reshape effort responses and act on their beliefs, almost optimally in all scenarios. Our results suggest a limited role for other-regarding preferences and can be rationalized by a selfish firm best responding to a worker evaluating wages according to a reference wage.

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1 Introduction

Ample experimental evidence convincingly demonstrates that other-regarding preferences lead people to deviate from the self-interested outcomes predicted by standard economic theory (Camerer, 2011). A significant stream in this literature models experimental labor markets under symmetric information and stable economic conditions to study how variation in structural parameters influences wages, effort, and employment (Charness and Kuhn, 2011). A general finding is that other-regarding preferences prop-up wages and effort above the Nash equilibrium prediction (Fehr et al., 1993, 1998), which motivates many behavioral labor market theories.[1]

However, the assumptions of symmetric information and stable economic conditions are unrealistic in most labor markets.[2] Firms often experience economic volatility and have discretion over how much information they share with workers about their financial conditions. These two factors can drive workers' effort response as they may accept wage cuts during well-known economic downturns (e.g., inflation in inputs or negative transitory shocks) but resist if unaware of the firms hardships (Bewley, 1995, 1999, 2007). Similarly, they may expect wage hikes during widespread prosperity and retaliate if unmet. When a firm's fortunes are private, a worker might not have enough information to demand higher compensation. Information asymmetries raise critical questions on the wage-effort relationship: on the one hand, how do firms adjust wages in response to economic volatility? And on the other, how do workers perceive wage changes based on available information? We propose that a worker evaluates wages with respect to a reference wage that depends on the firm's economic conditions.[3]

Prior research has separately examined economic instability (Davis et al., 2017; Buchanan and Houser, 2022; Bejarano et al., 2021) and information asymmetries (Rubin and Sheremeta, 2016; Dana et al., 2006, 2007; Feng et al., 2022; Werner, 2023; Buchanan et al., 2024) in principal-agent settings. Because of the relationship between reference wages and economic conditions, worker's knowledge plays a crucial role in wage-effort dynamics. Understanding this is important since

[1] Examples of theories include the fair wage-effort hypothesis (Akerlof, 1982; Akerlof and Yellen, 1988, 1990), reference dependence (Kahneman and Tversky, 1979), adverse selection in quits and hires (Weiss, 1980, 1990), and reciprocity and fairness (Rabin, 1993). For evidence on labor market dynamics, see Altonji and Devereux (2000); Agell and Lundborg (2003); Dickens et al. (2007); Babecky et al. (2010); Kaur (2019); Jo (2024).

[2] For example, using Danish data, Chan et al. (2020, 2023) report that wages adjust in the direction of productivity shocks and that adjustments are larger when productivity shocks are positive.

[3] This is consistent with recent evidence from Feng et al. (2022), which demonstrates that sellers behavior depends on reference points formed by both contracts and market price information.

most real-world employment relationships feature at least some degree of information asymmetry.[4] Furthermore, recent theoretical work suggests that these factors can potentially explain the sluggish wage adjustments and the dynamics observed in the U.S. labor market (Morales-Jiménez, 2022).

Despite the growing body of research on this topic, relatively little is known about how wage-effort relationships evolve when both changing economic conditions and information asymmetries are present. We address this gap by integrating permanent endowment shocks with information asymmetry while eliciting beliefs from both firms and workers regarding effort response functions and wage expectations. This design lets us investigate how economic and information conditions shape wages and effort and helps disentangle self-interest from fairness motives. For example, firms sharing gains even when workers are unaware suggests intrinsic fairness, whereas withholding gains under asymmetric information points to profit-maximizing behavior.

We use a 2×2 between-subjects design to study how economic and information conditions influence decisions in an experimental labor market. Specifically, we examine: (1) how firms and workers form and update beliefs, (2) how firms share economic gains and losses, and (3) the speed and symmetry of adjustments in wageeffort relationships. We match subjects into stable pairs and randomly assign one to act as a firm and the other as a worker and have them interact for 20 periods. In each period, the firm receives an endowment that it splits between the worker's wage and savings. The worker responds to the wage offer by choosing a costly level of effort. At the end of each period, both the firm and the worker learn their own payoff. Halfway through each session, we either increase or decrease (*positive* and *negative*) the firm's endowment by one-third. In some treatments, the workers know of this shock (*symmetric information*), and in others, they do not (*asymmetric information*). In the latter, workers receive no information hinting at the possibility of a shock, ensuring that from their perspective, economic conditions remain constant throughout the study.

Three key findings emerge from our experiment. First, comparing results across information conditions reveals that a considerable portion of gift exchange behavior is driven by self-interest rather than purely other-regarding preferences. Under symmetric information, wages adjust in the direction of endowment shocks, with firms and workers sharing shocks roughly equally. Our belief elicitation reveals that firms do this because they anticipate that informed workers expect wages

[4] See Cullen and Perez-Truglia (2018, 2022) for evidence of information frictions.

to adjust following endowment shocks. However, when workers lack information, firms capitalize on this asymmetry. Following a positive shock, firms keep a bigger portion to themselves and pass through wage increases that are only half as much as in the symmetric condition. Nonetheless, they still share the negative shock, with wages falling by about 20% less when firms know that workers are uninformed.

Second, we observe an asymmetric pattern in how workers update their expectations when facing wage surprises. When workers experience unexpected wage increases, they adjust their expectations relatively quickly –the gap between what they expect and what they receive vanishes after about 5 periods. In contrast, when workers face unexpected wage cuts, their expectations remain anchored to pre-shock levels. These misspecified expectations are costly for workers since we elicit them using accuracy-based incentives. As a consequence of the expectation mismatch, workers retaliate by choosing punitive effort levels that lower firms' payoffs.

Third, firms generally form accurate beliefs about workers' responses to wage changes, with one notable exception: they significantly underestimate workers' retaliation against unexpected wage cuts. This miscalibration explains why firms with an informational advantage in our negative asymmetric treatment end up worse off –they set wages below profit-maximizing levels, failing to fully anticipate workers' punitive responses.

Notably, these dynamics persist despite clear financial incentives for both parties to optimize their behavior: workers are incentivized to form accurate wage expectations, while firms are incentivized to maximize profits via wage-setting. Yet in the negative asymmetric shock scenario, we observe a persistent misalignment that causes sustained losses for both sides. This is not simply an information problem, as evidenced by the eventual convergence of actual and expected wages in the positive asymmetric information case. Rather, it suggests an asymmetry in how workers process and integrate negative shocks into market behavior, creating a 'belief wedge' that incentives alone fail to resolve.[5]

Our results highlight that worker behavior hinges on the economic conditions of the firm. Using

[5] This result aligns with Buchanan (2020) and Bushong and Gagnon-Bartsch (2024), who demonstrate that individuals project their current state onto others when predicting behavior, leading to systematic errors when others are in different states. Similarly, Buchanan et al. (2024) also find that firms' beliefs underestimate the effect of nominal wage cuts.

this insight, we calibrate a structural model where workers' reference wage depends on the firm's endowment. We then estimate our model using pre-shock data and forecast reactions in post-shock periods, benchmarking against a model with a constant reference wage. Our model significantly outperforms the benchmark in all measures, demonstrating the value of adopting a dynamic reference wage that depends on the firm's conditions.

Our work intersects with and extends experimental literature on how economic shocks influence gift exchange.[6] Previous studies have looked at the effect of exogenous random shocks affecting either workers' productivity or firm conditions. Gerhards and Heinz (2017) investigate how the anticipation of adverse shocks impacts gift exchange by incorporating probabilistic exogenous shocks into a two-period gift exchange game. They show that the mere possibility of a shock led to higher initial wages, which would subsequently decrease if the shock occurred, yet workers maintained reciprocity.[7] In a related work, Bejarano et al. (2021) introduce probabilistic productivity shocks, both positive and negative, showing that wage changes occur under economic instability in general. Several related studies include firm-side or market-wide exogenous shocks. Buchanan and Houser (2022) shed light on the effect of recessions in a standard gift-exchange game and find that wage cuts lead to lower workers effort. We add to this literature by shedding light on the role of information asymmetries under economic unstable periods. While we corroborate these results under symmetric information, we show that the presence of asymmetric information exacerbates firms tendencies to behave opportunistically. Furthermore, relative to previous studies, we directly elicit incentivized beliefs from both firms and workers rather than inferring reference wages from past wages as in Bejarano et al. (2021). This helps us to understand how firms anticipate workers' effort reactions and exploit information frictions to their advantage. In addition, we show that workers' reference wage also depends on the firm's conditions and not solely on past wages. Hence, they adjust to new information about the firm.

Our work also contributes to prior experimental studies on the effect of information asymmetries.[8]

[6] There is an extensive literature modeling experimental labor markets as gift-exchanges. Examples using chosen effort are Fehr et al. (1993); Fehr et al. (1998) and Charness (2004). Examples using real-effort tasks are Cohn et al. (2015); Greiner et al. (2011) and Hennig-Schmidt et al. (2010).

[7] Other relevant studies show that temporary, probabilistic tax shocks that hit wages, productivity, or both can mute gift exchange following a wage shock (Kujansuu and Schram, 2021) or, on the contraty, gift exchange remained robust to cyclical productivity changes (Kocher and Strasser, 2011).

[8] There is some evidence from dictator and ultimatum games which suggests that people strategically exploit asymmetric information in their favor when making pro-social decisions (Dana et al., 2006, 2007) and that varying

Prior studies provided evidence that wage cuts were viewed as unfair unless rationalized by known exogenous conditions, like an economic downturn (Kahneman et al., 1986; Chen and Horton, 2016). Our study relates to this strand of literature by reporting evidence that unjustified wage cuts lead to severe worker retaliation, while justified cuts following observable shocks are punished less harshly. Closely related to our study is Rubin and Sheremeta (2016) who show that random productivity shocks impede gift exchange even when firms are fully informed about the shock.[9] These findings imply that the imprecision of output metrics in organizations will tend to weaken gift-exchange. We add to these findings by i) studying firms' behavior when facing instability, rather than shocks affecting workers' productivity, and ii) that changing economic conditions tend to weaken gift-exchange even in the case in which workers are not perfectly informed about firms' economic conditions.

Our work offers insights to the literature on reference dependence (Kőszegi and Rabin, 2006; Crawford and Meng, 2011; Abeler et al., 2011). Koch (2021) finds that contracts act as reference points, causing slightly more rigid wages during negative profit shocks in a symmetric information setting where firms and workers both know effort levels and economic conditions. Similarly, Doerrenberg et al. (2023) demonstrate that online workers respond more strongly along the extensive margin to wage cuts than to wage hikes. We contribute to this literature in several ways. First, we show how restricting information to workers leads to more muted and sluggish adjustments of reference wages. Second, we study wage-effort dynamics in response to both expansionary and recessionary shocks under various information conditions. We find an asymmetry in experience-based adjustments: workers update more rapidly to wage increases than to cuts. In contrast, news-based adjustments are immediate and symmetric across conditions.

We offer insights into the growing literature on downward wage rigidities, inspired by the seminal works by Bewley (1995, 1999) and empirically tested in Kaur (2019), Dickson and Fongoni (2019) and Fongoni et al. (2024). Our work demonstrates how information asymmetries and reference-dependent wage expectations contribute to the empirical fact of wage-cut resistance, even in environments in which firms face economic shocks and have incentives to adjust wages. Additionally, our contribution is also methodological. Our extensive belief elicitation allows us to show that firm be-

information in ultimatum games shows that relative payoffs and fairness enter players' utility functions (Mitzkewitz and Nagel, 1993; Kagel et al., 1996; Schmitt, 2004; Croson, 1996).

[9] Yet, Davis et al. (2017) do not fully replicate these findings, suggesting that information about shocks to firms can restore the gift exchange relation.

liefs about worker effort provide an additional mechanism to explain firms' reluctance to cut wages, consistent with survey evidence from Bewley (1995, 1999). Our results also align with Kahneman et al. (1986), who show that wage cuts are deemed fair when a firm faces financial distress but are resisted when perceived as opportunistic. In our experiment, workers accept wage cuts when aware of an exogenous shock but punish firms that attempt the same cut without transparency.

Section 2 briefly describes our conceptual framework, highlighting the role of information asymmetries, and describes our experimental design. Section 3 contains our main analysis and results. We show the results of our structural exercise in section 4. Finally, section 5 concludes.

2 Experimental Design

Our conceptual framework illustrates how information asymmetries and economic shocks affect wage-effort dynamics. We model workers who care about both the absolute level of wages and their relation to the firm's economic condition (i.e., the firm's endowment). For simplicity, we consider a static framework where neither past experience nor reputation plays any role, although we will explore these factors in our main analysis.[10] The key intuition is that workers' reference-dependent preferences are calibrated relative to the firm's endowment. Consequently, changes in the endowment will reshape the effort response function – but *only if* workers are aware of these changes. Previous models employ similar logic by allowing for effort to respond to the introduction of minimal wages (Brandts and Charness, 2004; Falk et al., 2006; Owens and Kagel, 2010), a rise in inflation levels or a threat to firm profits (Kahneman et al., 1986; Kaur, 2019).

2.1 Conceptual Framework

Consider a labor market with a single firm and single worker formed into a firm-worker pair. The firm has an endowment R which she uses to set a wage $w \in [\underline{w}, ..., R]$. The worker provides costly effort $e \in [0, \overline{e}(w)]$, with $c(e)$ being the cost of effort and $\overline{e}(w)$ is the maximum possible effort that yields non-negative monetary payoffs (i.e, $w - c(\overline{e}) = 0$). For now, we assume that both the worker and firm have symmetric information regarding economic conditions, captured by R.

We adopt a widely-used profit function, first introduced in Fehr et al. (1993) and Fehr and Schmidt

[10] An interesting expansion of the model is that of incorporating the fact that sudden wage changes may signal to workers a change in firms' economic conditions. While our actual design uses a dynamic setting so that we can study belief updating, we do not model these situations and limit the aim of this simple static model to develop intuition. Comparative statics help mimic average pre- and post-shock behavior in our experimental setting.

(1997), that models a profit-maximizing firm with no altruistic motives whose profit function depends on the endowment, the wage, and the worker's effort.

$$\pi_i(w, e) = (R - w)e. \tag{1}$$

We show in Appendix B that our hypotheses are robust to assuming an alternative profit function, $\pi = Re - w$.[11] The worker's utility function incorporates monetary incentives, reference-dependence, and other-regarding preferences. This yields a modified version of the effort response function introduced in Akerlof and Yellen (1988, 1990) which is similar in spirit to Dickson and Fongoni (2019):

$$U_j\left(e, w, \tilde{R}\right) = w - c(e) + M\left(w, \tilde{R}, w_0(\tilde{R})\right) \cdot e \tag{2}$$

where $c(e)$ is a strictly convex cost function of effort. The first component $w - c(e)$ captures the monetary incentive of the worker while the last component M is the 'morale function,' which captures three distinct features comprising the worker's reference dependence and other-regarding preferences: w is the wage offer, $\tilde{R}$ is the belief of the firm's economic conditions, and $w_0(\cdot)$ is the reference wage that depends on her beliefs about the firm. Since we assume $c(e) = .5e^2$ in our experiment, the optimal effort level is $e = M\left(w, \tilde{R}, w_0(\tilde{R})\right)$. Hence, how the worker reacts to economic shocks and economic frictions critically depends on the assumptions we make on the moral function. Based on those, we can then study how firms react to shocks and set wages.

2.2 Endowment shocks and the role of information frictions

We assume a functional form for the moral function M and solve the model via best response analysis (details in Appendix B), yielding the following optimal effort and wage levels:

$$e_i^*\left(w, \tilde{R}, w_0(\tilde{R})\right) = \alpha_j \left(\frac{\mu + w - w_0(\tilde{R})}{\tilde{R}}\right)^{\beta_j} \tag{3}$$

$$w^*(e^*) = \frac{\beta_j R + w_0(\tilde{R}) - \mu}{1 + \beta_j} \tag{4}$$

[11] The advantage of this form is that the marginal product of the effort is not decreasing in the wage. However, it has the caveat of possible negative payoffs. Hence, we decided not to implement it in our experimental design. Another alternative we have encountered is $\pi = R - we$. However, with this functional form, profits are decreasing in the effort, and a firm could maximize profits by simply offering a wage of zero. Further, this solution is not unique. Imagine instead that the firm pays a wage of R. If the worker responds with $e = 0$ then the firm still earns $R - R * 0 = R$.

Our functional form assumes $w_0(\cdot)$ is order-preserving, so a higher endowment always warrants a higher wage. At the same time, the worker considers wages relative to the firm's endowment. For example, suppose a worker receives a wage larger than her reference wage. Intuitively, our model says the worker will more strongly reward this wage coming from a poorer firm than a richer firm. This adds an additional channel through which changes in (information about) economic conditions impact the worker's effort choices.

Using both equations 3 and 4, we can now illustrate the dynamics of the model in response to an endowment shock under both symmetric and asymmetric information conditions. Note that since firms care about workers' responses and workers care about fairness, the firm indirectly incorporates fairness concerns when maximizing profits via wage selection.

We consider only exogenous shocks, not based on the firm decisions.[12] Suppose there is an exogenous expansionary (recessionary) increasing R to R' that is known to both the firm and the worker. This shock puts upward (downward) pressure on wages via R and $w_0(R)$. Thus, both the worker's reference wage and the optimal wage adjust in the direction of an endowment shock under symmetric information. In this scenario, a change in R changes the optimal wage $w^*(e^*)$ by $\frac{\partial w^*}{\partial R} = \frac{\beta_j + w_0'(R)}{1+\beta_j}$.

Hypothesis 1: Under symmetric information, a positive (negative) shock to the firm's endowment leads to an increase (decrease) in the wage offered by the firm.

Now let us consider the worker's effort response to a given wage w. Changes in the endowment R impact effort levels through two channels: $w_0(R)$ and the scaling factor R. When R increases, both imply a decrease in the effort level for any given wage. This yields the following hypothesis:

Hypothesis 2: Under symmetric information, a positive (negative) shock to the firm's endowment leads to the worker providing less (more) effort, relative to pre-shock effort levels, for each wage in the wage distribution.

Suppose instead that only the firm knows about the exogenous expansionary shock. Firm's endowment shifts from R to R', leading to an increase in $w^*(\cdot)$ directly through the change in the endowment. However, the worker does not adjust her belief about R, $\tilde{R}$, to its new level so that

[12]Our experimental setting best captures the circumstance of a firm facing an exogenous economic shock. Workers' responses to idiosyncratic, endogenous shocks could differ. For example, if shocks were due to either good or bad management. However, we leave that heterogeneity outside our model and for future research.

$R = \tilde{R} < R'$. As a result, $w_0(R) = w_0(\tilde{R}) < w_0(R')$. Since equation 4 is strictly increasing in $w_0(\cdot)$, the optimal wage w^* under asymmetric information is lower that its analog under symmetric information.

Hypothesis 3: Information frictions moderate wage changes following an endowment shock.

When the worker is unaware of the endowment shock, her effort response function remains unchanged, $R = \tilde{R}$. Because of this, the worker will react more strongly to a given wage increase than how she would react to the same wage increase under symmetric information since $\frac{w-w_0(R)}{R} < \frac{w-w_0(R')}{R'}$ when $R' > R$.[13] This leads to the following hypothesis:

Hypothesis 4: Workers respond more strongly to wage changes following an endowment shock under asymmetric information than symmetric information.

Overall, this simple model illustrates that information structure can play a critical role in determining allocations following economic shocks. Our experimental design, which closely mimics the intuition of this model, provides a direct test for these hypotheses. While we do not allow firms to strategically disclose endowment shocks, another prediction of our model is that a firm will always want to reveal negative news. However, other considerations outside the scope of our simple model such as outside investment, reputation or stock prices might cause firms to not disclose negative news. Though including these features could lead to interesting insights, we leave these nuances for future work.

2.3 Experimental Design

Our experimental design extends the classic gift-exchange game in three important ways. First, firms face either a positive or negative permanent shock to their endowment. Second, we introduce information frictions by varying workers' awareness of the shock. Finally, we elicit firms' beliefs and workers' effort strategies both before and after the shock occurs.[14] The confluence of these extensions allows us to distinguish between self-interest and other-regarding motives in firms' wage choices and to understand the role of information frictions in wage-effort dynamics.

The firm receives an endowment of R at the beginning of each period and makes a wage offer w

[13] This assumes $\alpha_j > 0$, which is akin to assuming at least some degree of other-regarding behavior.
[14] Armouti-Hansen et al. (2020) use a similar elicitation method in their recent working paper.

using a simple input box. We inform the worker of w, then allow the worker to choose costly effort $e \in \{0, ..., \overline{e}(w)\}$, where $\overline{e}(w)$ is the largest effort level that does not yield a negative payoff for the worker. We provide the worker with a slider tool that displays payoff information for both the firm and worker based on hypothetical effort levels to help them make this choice. In asymmetric treatments, we conceal from the worker the firm's payoff information, as this would reveal the shock. Hence, workers choose their effort levels without this extra information. Although this difference could, in principle, introduce bias between conditions, we find no significant differences in effort levels across treatments during the pre-shock periods. This suggests that the absence of payoff information did not affect workers' effort choices. We provide examples of both the firm's and the worker's decision screens in the Appendix (see Appendix section F). Both players then receive feedback on their respective payoffs and the firm also learns about the worker's effort.

After 10 rounds of baseline play where $R = 12$, we introduce a permanent session-level endowment shock: $R = 16$ following a positive shock and $R = 8$ following a negative shock. Shocks are common knowledge in the symmetric information treatment. However, only firms learn of the shock in the asymmetric information treatment, and workers have no possibility to learn or infer the change in endowment. The only exception is firms that offer wages higher than 12 in the positive asymmetric condition, which in our sample occurs only twice. In those instances, we reveal the shock to that particular worker in the same fashion as in the symmetric information treatments. To ensure that firms understand the information structure in our asymmetric information treatments, we add a short comprehension test regarding firms' beliefs about workers' knowledge of the shock.[15] Variation in these two factors, shock direction and information structure, yields our 2×2 between-subjects design summarized in Table 1.

We elicit the firm's beliefs of her own worker's effort response, and her worker's corresponding effort response function by implementing the strategy method in periods 5, 10, 11, and 15. This allows us to understand how beliefs evolve as a function of experience during baseline play, how beliefs respond to shocks, and how the structure of information surrounding the shock affects beliefs. We do this by asking what effort the participant playing the role of firm expects from the worker at each possible wage.

[15] We screen out confused subjects using results from this quiz. We find no differences in confusion across treatment conditions (6 subjects in positive shock and 4 in negative shock, p-value=0.267).

Table 1: Summary of Design Treatments

Treatments	Shock Information	Endowment	Participants
Positive	Firms and Workers	$12 \rightarrow 16$	56
Negative	Firms and Workers	$12 \rightarrow 8$	66
Pos. Asymmetric	Only Firms	$12 \rightarrow 16$	56
Neg. Asymmetric	Only Firms	$12 \rightarrow 8$	72

In these same periods, we also elicit the worker's wage expectation for the current period and use the strategy method to elicit the worker's corresponding effort response function after first asking her wage expectation in that period. We provide example screenshots for both in Appendix F.

We incentivize the worker during elicitation periods by implementing her effort response function and paying her $5 for a correct wage belief. We incentivize the firm's beliefs by paying $5 if the effort guess of a randomly selected wage from the wage distribution is within .25 units of the actual effort response for that wage.

During elicitation periods, both the firm and the worker can explore all hypothetical choices before providing a response. Similar to how the worker chooses effort in non-elicitation periods, both the firm and the worker can use a slider to understand how a potential effort level for each wage translate into payoffs. After the first elicitation period, we remind each subject of their own choices in the previous elicitation period. Thus, any changes in beliefs or effort reflect actual updates and not a lack of recollection. We provide instructions on screen for subjects during all elicitations.

Before moving to results, we briefly address two possible design concerns. First, we circumvent hedging in elicitation periods by randomly paying subjects for either the accuracy of beliefs or for choices.[16] One may wonder if employing the strategy method during elicitation periods induces an artificially high degree of monotonicity in effort responses. However, Maximiano et al. (2007) shows in a multi-worker gift-exchange game that the strategy method has a negligible effect, if any, on subject behavior. This finding is especially true in low-complexity environments.[17]

[16] For example, a firm that suspects a worker will not reciprocate but wants to offer a high wage with the hope of reciprocation may guess that the worker will provide low effort at this wage.

[17] See Cason and Mui (1998); Brandts and Charness (2000); Oxoby and McLeish (2004); Bosch-Domènech and Silvestre (2005).

Second, we consider our specific choice of information structure. Despite information frictions, participants are always aware of how their choices will impact their per-period monetary payoffs in all treatments and roles. Although workers sometimes have incomplete information since they do not observe the firm's payoff, this does not constitute deception according to the long-standing convention about information omission in experiments (Hey, 1998; Hertwig and Ortmann, 2008; Charness et al., 2022). As a robustness test, we run a separate experiment with a smaller number of subject pairs in which we explicitly reveal to workers that endowments might change during the experiment, introducing the possibility of higher-order beliefs to influence wage and effort dynamics. For example, a firm could exploit this alternative design by introducing a discrete wage cut, hoping the worker rationalizes the wage cut as the result of an exogenous shock to their endowment. We find that our results are robust to this alternative (see details in Appendix E). This is consistent with an exit survey, where half of the participants do not think a shock occurred, and only 30% guess the direction of the shock correctly.

2.4 Procedures and Implementation

We recruited a total of 250 undergraduate student subjects from Texas A&M University via the recruitment system ORSEE (Greiner, 2015). We programmed all experimental software using oTree (Chen et al., 2016).

Each session began by reading experimental instructions aloud and administering comprehension quizzes. We deliver instructions in two parts. First, we provided subjects a brief set of instructions outlining basic components of the game like experimental length, formation of pairs, and anonymity (see instructions in Appendix D.1), and then administered a first comprehension quiz. We then provided a second set of instructions detailing game play and payoffs (see instructions in Appendix D.2) followed by a second comprehension quiz.

Following the second quiz, we randomly assign participants to the role of either firm or worker and form firm-worker pairs and familiarize them with the experiment via three unpaid practice periods. Following practice periods, we randomly form new firm-worker pairs that remain stable for the subsequent 20 paid periods. Subjects completed a short socio-demographic survey and the end of the experiment. Following this survey, we released subjects and paid them privately. The average payment in our sessions was $25 and each session lasted approximately 90 minutes.

3 Results

We find that information frictions play a fundamental role in determining the allocation of exogenous economic shocks in our experimental labor markets. Under symmetric information, shocks are shared almost equally between firms and their counterparts, while information frictions produce an asymmetric response, with uninformed workers retaliating against wage cuts. A key outcome is that the more informed party does not always benefit from information frictions.

Firms' beliefs are highly accurate in most treatments and they rapidly adjust to economic shocks. Workers' wage expectations also adjust to information about economic shocks. However, in the absence of information and relying only on experience, we find a sluggish and asymmetric adjustment in expectations following economic shock

3.1 Wage-Effort Dynamics in symmetric information

We first consider wage-effort dynamics. Figures E8c and E8d show that both wages and effort adjust as a consequence of the endowment shocks. Under symmetric information, average wages increase by about 2.25 units following a positive exogenous economic shock. This wage increase – slightly more than half of the shock – yields a small, albeit statistically insignificant, response in effort (p-value = 0.229). Negative shocks induce an average wage cut of 2.1 units, which again prompts a small, statistically insignificant decrease in effort (p-value = 0.157). Both wage changes are highly significant (p-values < 0.001), in line with Hypothesis 1, and of similar magnitudes, indicating a symmetric response to shocks.

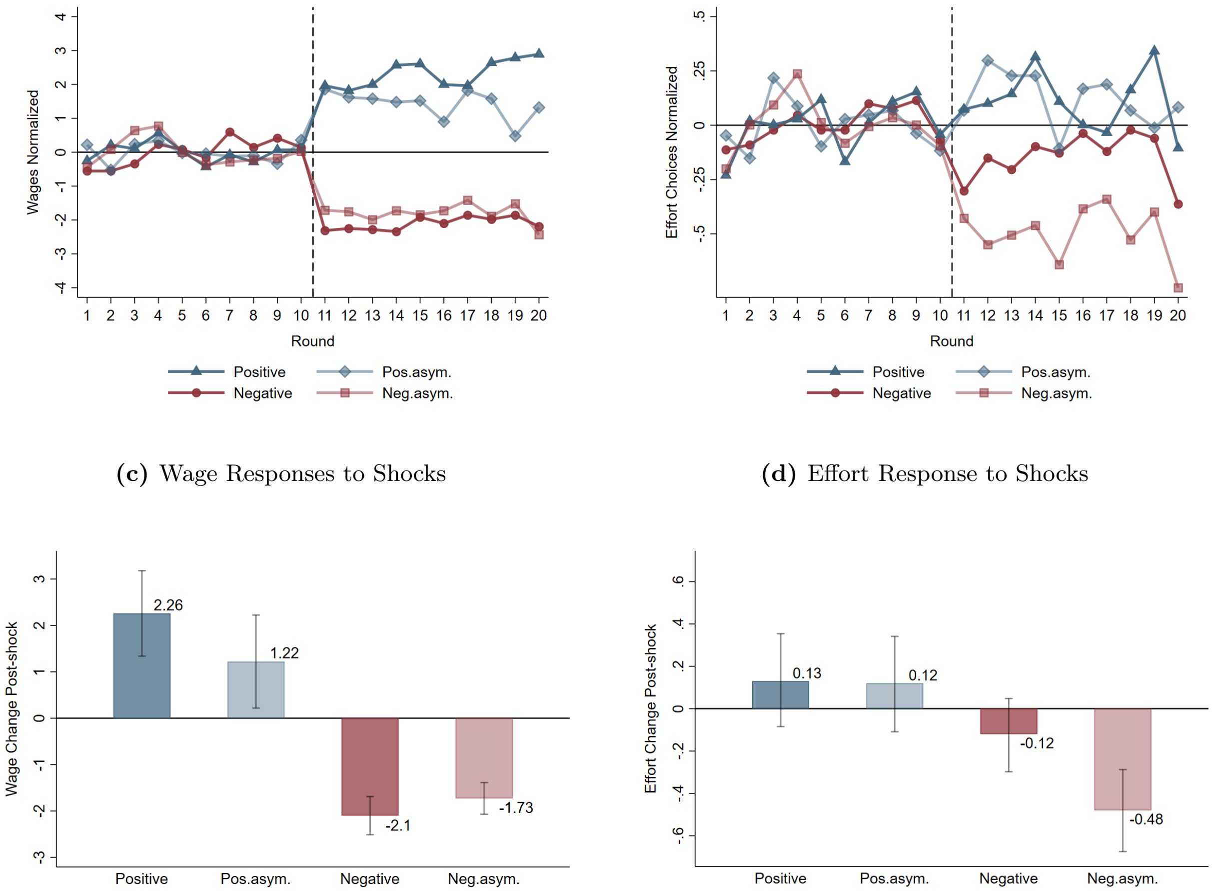
> **Result 1:** *Under symmetric information, wage adjustments are positively correlated with endowment shocks but do not induce significant changes in effort.*

These changes in wages and effort are consistent with firm-level beliefs about workers' effort response functions. To see this, consider Figures 2a and 2b, which show how endowment shocks reshape firms' beliefs about worker's responses. Firms' choices reveal that, under symmetric information, firms' beliefs move in the opposite direction of the endowment shock: they expect less effort for a given wage after a positive shock, indicated by a downward slope in the pre/post beliefs, and more effort following a negative shock, which causes the upward slope. Put another way, firms in our symmetric information treatments are obligated to share a positive shock to maintain effort levels and are allowed to share a negative shock without reprisal from the worker.

Figure 1: Average Wage and Effort Responses by treatment

(a) Wage offers **(b)** Effort responses

(c) Wage Responses to Shocks **(d)** Effort Response to Shocks



Notes: This figure shows the average wage (a) and effort choice (b) for each round. We normalize each variable by subtracting the pre-shock average and exclude pairs where the firm did not pass the asymmetric test. We depict the average change in wages (c) and effort (d) following endowment shocks in each of our four treatments (bars) surrounded by 95% confidence intervals estimated using a series of random-effects linear regressions that group pre- and post-shock data. We control for the final round, gender, if the pair played the Nash equilibrium strategy (defined as the worker selecting effort equal to 0 for all possible wages in at least one elicitation) and for the corresponding average wage offered during the trial rounds. Clustered standard errors at the individual level. We report the full output in Tables A6 and A7.

Firms' beliefs about how shocks impact effort responses are quite accurate (see figures A5 and A6). This suggests that firms in our experiment correctly anticipated that workers care about the intentions behind wage choices. Notably, this holds even when firms predicted effort changes at wage levels they had not previously offered.

Result 2: *Under symmetric information, firms' beliefs reflect workers' expectations of sharing the consequences of exogenous economic shocks, regardless of direction. These beliefs are accurate.*

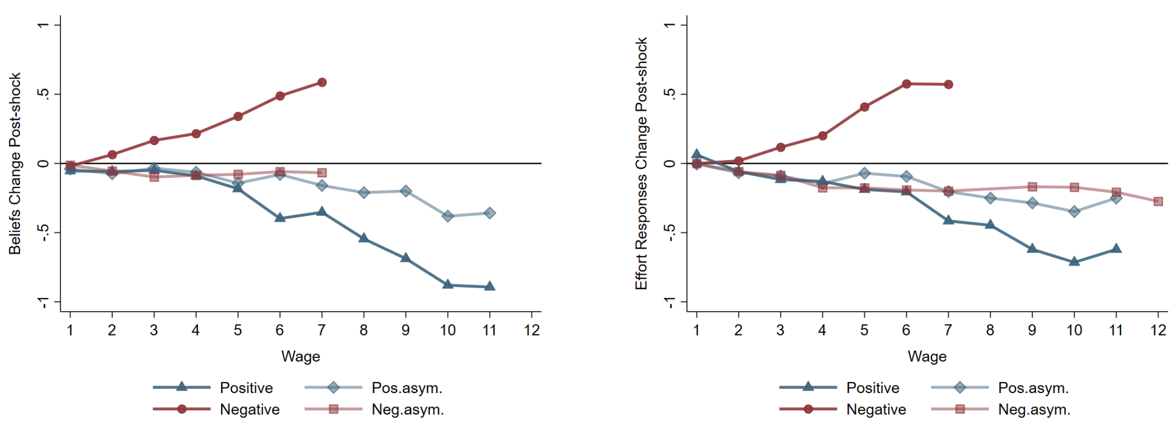
The fact that the effort-response function adjusts in Figure (2b) following these shocks is consistent with our theoretical framework and supports Hypothesis 2. This suggests that workers consider more than just the level wage level when making effort decisions. Instead, our results suggest that workers care about the wage relative to the firm's endowment. Firms anticipate this and adjust wages accordingly following endowment shocks, which supports the assumption of accurate contingent reasoning in both classical and behavioral models of principal-agent interactions.

Result 3: *Workers' effort responses to a given wage account for the economic conditions of the firm. This is reflected across the full wage schedule.*

Figure 2: Beliefs and Effort Responses to Shocks

(a) Pre/Post Effort Beliefs Change

(b) Pre/Post Effort Responses Change



Notes: This figure depicts the average change in firms' beliefs and workers' effort responses between pre and post-shock periods in each of our four treatments. Pre-shocks elicitations are pooled together, similarly with post-shocks elicitations. A positive (negative) slope indicates an increase (decrease) in expected effort with respect to the pre-shock period. We excluded pairs in the asymmetric information conditions where the firm did not pass the asymmetric info check. For more details, refer to Table A8 and A9. In addition, we report non-pooled beliefs and effort responses in Figures A5 and A6.

3.2 Wage-Effort Dynamics in Asymmetric Information

Information frictions moderate the reaction of wages to endowment shocks and also impacts how workers respond to wage cuts. These results provide support for Hypothesis 3. Wages increase by only 1.22 (p-value $= 0.017$) units on average following the positive shock, which is a bit less than half of the wage hike following an equivalent endowment shock under symmetric information (diff. 1.13, p-value $= 0.092$). Workers do not exhibit a significant response to this muted wage hike. Following a negative shock, firms cut wages by 1.7 units, which is about 20% less severe than the wage cut under symmetric information, although it is statistically similar (p-value $= 0.258$). Despite this smaller wage cut, workers exhibit a large and highly significant reduction in effort of .48 units (p-value < 0.001). This effort cut is four times larger than the average effort cut under symmetric information.

Hypothesis 4 states that workers would be more responsive to wage changes under information frictions. This is clearly true following negative shocks, where effort cuts are four times as large under information frictions even though wages fall by 20% less. The positive shock case is more nuanced. In the positive treatments, although wage increases are half as large, they induce the same effort change from the workers. In other words, firms can increase wages by half as much to induce a similar effort response under information frictions.

> **Result 4:** *Information frictions moderate wage dynamics but amplify the wage elasticity of effort following shocks. This amplification is asymmetric; effort responds much more strongly to wage cuts than hikes.*

Firm beliefs about effort response in 2a reflect a clear understanding that information frictions conceal shocks from workers. Whereas firms expect shocks to reshape the effort response function under symmetric information, firms correctly predict a static effort response function under information frictions. This difference in beliefs, coupled with choice data, gives critical insight into the motives underlying firm behavior.

What we see is that firms facing full-information conditions correctly believe that workers will demand higher wages and thus, they pay higher wages. If firms were solely other regarding then we would expect equivalently high wage hikes following a positive shock under both information conditions. Instead, we see that the firms who believe workers will not demand higher wages do not pass through of positive shocks. Taken together, this suggests a self-interested firm that increases wages under symmetric information partly because they know the worker will demand higher wages.

What we see is that firms facing full-information conditions correctly anticipate how workers' effort functions respond to shocks and adjust wages accordingly. We also observe that firms correctly anticipate that uninformed workers will not demand higher wages following a positive endowment shock and will not tolerate wage cuts as severe as those under full information following negative shocks. If other-regarding preferences alone motivated how firms allocate shocks, we would expect similar wage increases following a positive shock under both information conditions. Instead, we see that firms who believe workers will not demand higher wages do not fully pass through positive shocks. This pattern suggests that firms increase wages under symmetric information primarily because they anticipate workers' economic demands, not solely due to fairness considerations.

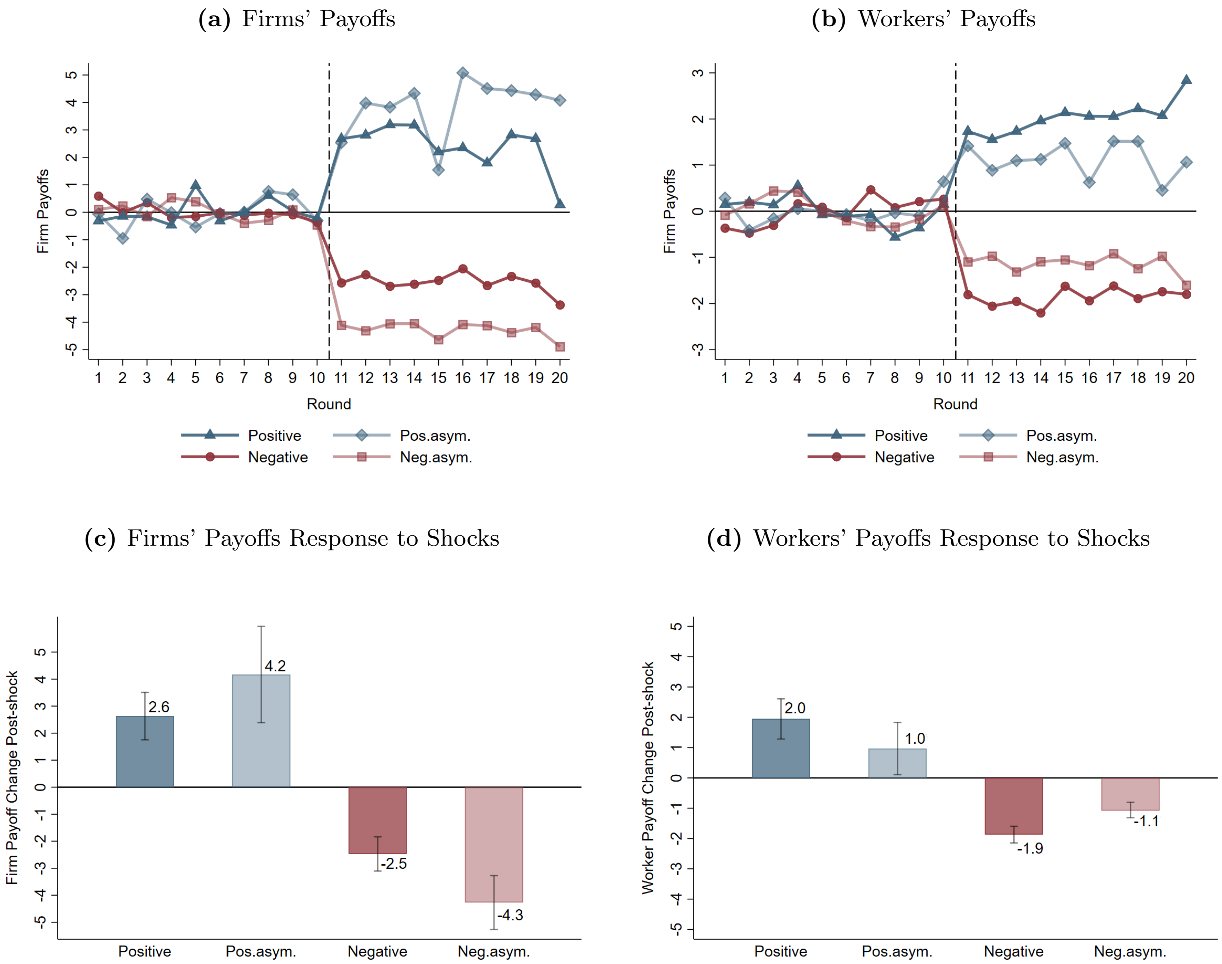
Result 5: *Other-regarding preferences alone cannot explain how firms share positive shocks, since wage increases are only about half as large in asymmetric information than in symmetric information.*

3.3 Payoffs and the Distribution of Endowment Shocks

Our primary finding in this subsection is that information structure matters for determining allocations of shocks. Interestingly, it isn't always the more informed party who benefits from information frictions in our experiment. To see this, consider Figure 3, which shows two things. First, information frictions benefit firms following a positive shock and harm firms following a negative shock. Second, the opposite is true for workers, whose payoffs increase less following positive shocks but fall less following negative shocks whenever information is asymmetric.

Under symmetric information, firms and workers split endowment shocks almost equally. Because of this, positive shocks increase payoffs for both firms and workers by a similar amount (2.6 vs 2 units, p-value= 0.216), and negative endowment shocks decrease payoffs by comparable magnitudes (2.5 vs 1.9 units, p-value= 0.084).

Figure 3: Firms and Workers Payoffs Changes to Shocks



Notes: This figure shows the average firms' (a) and workers' payoffs (b) for each round. We normalize each variable by subtracting the pre-shock average and exclude pairs where the firm did not pass the asymmetric test. We depict the average payoffs for firms (c) and workers (d) in each of our four treatments (bars) surrounded by 95% confidence intervals estimated using a series of random-effects linear regressions that group pre- and post-shock data. We control for the final round, gender and if the pair played the Nash equilibrium strategy (defined as the worker selecting effort equal to 0 for all possible wages in at least one elicitation). We excluded pairs where the firm did not pass the asymmetric info check. Clustered standard errors at the individual level. We report the full output in Tables A10 and A11 in the Appendix.

Making information about shocks asymmetric greatly benefits firms following positive endowment shocks. Information frictions allow firms to capture a much larger share of the shock without negatively impacting worker effort. This is because workers in asymmetric information treatments do not adjust their reference wage following the shock, which implies that their effort response functions remain unchanged (see Appendix, Figure A6).

Firms correctly predict this (Figure 2a), and significantly reduce pass-thru of the positive endowment shock. Though wages increase less when information is asymmetric, effort responses are equivalent to the symmetric information setting. This causes firm profits, on average, to increase by about twice as much relative to the symmetric information setting (1.85 units increase, p-value = 0.063). In contrast, workers' payoffs increase little - if at all - when information is asymmetric. Taken together, this means firms increase their profits by about four times as much as workers following a positive shock.

Information frictions harm firms following negative endowment shocks. The average firm's payoff falls by about 72% more under asymmetric information (1.77 units decrease, p-value = 0.003). Though firms correctly infer that the worker's stated effort response function is static across the shock due to asymmetric information (Figure 2a), they share the shock nonetheless.[18] Wage cuts cause wages to fall significantly below the worker's reference wage, which were also static due to asymmetric information. This leads to drastic and punitive effort cuts that severely impact firm profits.

On the other hand, workers benefit from information frictions following a negative shock. Although they are the less informed party, the average worker payoff falls by only about 57% of what it does under symmetric information (0.73 units increase, p-value < 0.001). This happens because punitive effort cuts greatly reduce the cost of effort. Coupled with a slightly higher average wage, this leads to a higher payoff relative to the symmetric information setting. However, asymmetric information is less beneficial to workers when the endowment shock is positive; average payoffs increase by about half of what they do under symmetric information (1.09 units decrease, p-value = 0.044).

Result 6: *Under symmetric information, profits adjust equally for firms and workers regardless of shock direction. Under asymmetric information, it is not always the more informed party that benefits from information frictions. Firms capture significantly more profit following a positive shock but experience larger profit losses following a negative shock.*

[18] This aligns with Equation 4 where wages always adjust in the direction of endowment shocks.

3.4 Optimal Wages and Beliefs Accuracy

After studying wage-effort dynamics, we next look at whether firms form accurate beliefs about optimal wages and if they act on these beliefs. Overall, we find that firms act as profit maximizers: they do form accurate beliefs and make beliefs-consistent decisions. This provides suggestive support that people adhere to the sort of best-response analysis commonly used to solve principal-agent problems in theory.

Table 2 reports three key pieces of information. We report in column 1 the wage that maximizes firm profits based on firms' beliefs about workers' effort response functions. Column 2 reports the wage that will actually optimize firm profits based on workers' elicited effort responses. Finally, column 3 reports the wage that a firm actually offers.

Firms' beliefs are mostly accurate: optimal wages based on firms' beliefs (column 1) closely match optimal wages based on workers' effort responses (column 2) in all treatments prior to any endowment shock. And although endowment shocks shift effort responses, firms anticipate this and adjust wages accordingly so that offers never diverge significantly from the actual optimal wage.[19] The one exception we observe comes in our negative asymmetric information treatment, where firms do not anticipate workers retaliating for unexpected wage cuts with punitive effort cuts. Thus, the average firm's offer is too low relative to the actual average profit-maximizing wage.

Firms' actions are consistent with their beliefs. Comparing columns 1 and 3 of Table 2 confirms that firms offer wages that are optimal according to their beliefs. Differences between these columns are never statistically significant. This is true even in our one treatment (negative asymmetric) where the firm holds incorrect post-shock beliefs about the profit-maximizing wage.

Result 7: *Firms typically form highly-accurate beliefs. However, firm beliefs are misspecified following negative asymmetric shocks. Firms act on their beliefs, including their misspecified beliefs. This leads to a belief-driven failure in profit maximization.*

[19] Note that even if beliefs and effort responses do not change, optimal wages are still impacted by changes in endowment, as seen in equation 4.

Table 2: Expectations and wage optimality rounds 5, 10, 11 and 15

	(1) Optimal wage wrt beliefs	(2) Optimal wage wrt responses	(3) Wage offer	(1) - (3) Difference p-value	(2) - (3) Difference p-value
Pre (+)	6.35	6.25	6.66	0.48	0.22
Post (+)	8.53	8.62	8.89	0.50	0.65
Pre (+Asym.)	6.04	6.40	5.97	0.88	0.38
Post (+Asym.)	6.61	7.06	7.31	0.15	0.75
Pre (-)	6.81	7.46	6.81	1.0	0.12
Post (-)	4.56	4.96	4.59	0.91	0.12
Pre (-Asym.)	5.54	6.23	5.68	0.70	0.18
Post (-Asym.)	3.89	4.95	3.93	0.87	0.04

Notes: Optimal wage with respect to beliefs is the average of optimal wages according to firms beliefs. Similarly, Optimal wage with respect to responses is the average of the optimal wages based on workers response function. Wage offer is the wage offered by the firm in the elicitation rounds. We exclude pairs where the firm did not pass the asymmetric test. Pre-shock and post-shock rounds are pooled respectively. Columns 4 and 5 report the p-values of a paired t-test for the equality of columns 1-3 and 2-3, respectively.

3.5 Dynamic Reference Dependence and Information Frictions

We now consider how workers' wage expectations adjust following endowment shocks.[20] Two key insights emerge. First, we find that wage expectations adjust instantaneously and accurately in response to information about economic shocks, which aligns with Abeler et al. (2011).[21] Second, when information is asymmetric, wage expectations can still adjust based on post-shock experience (in line with Thakral and Tô, 2021). However, adjustment is sluggish relative to a full-information setting and adjustment dynamics depend on the direction of wage surprises.

Under symmetric information, workers' wage expectations are accurate in both pre- and post-shock periods. These expectations are stable on either side of the shock but adjust between periods 10 and 11. We show this in Table 3, which reports average wages, average wage expectations, and their difference for each elicitation period in all treatments. This is true even though workers have not experienced any post-shock wage offers, which highlights the importance of information about the

[20] One can think of wage expectations as a reference point, akin to Kőszegi and Rabin (2006).

[21] Abeler et al. (2011) manipulate the rational expectations of subjects and show this impacts wage-effort dynamics via a reference wage. We show that information about changes in economic conditions, coupled with previous worker-firm experience, is enough to instantly shift wage expectations.

firm's endowment in establishing wage expectations and effort responses. The stability of wages and expectations between rounds 5 and 10 allay any concern that the updating in wage expectations between rounds 10 and 11 are a continuation of pre-shock trends.

Discrepancies occur in asymmetric information treatments because information frictions prevent wage expectations from adjusting to the real endowment. Since wages adjust in the direction of endowment shocks, workers receive an unexpected wage hike following a positive shock and an unexpected wage cut following a negative shock.

We find that experience alone in the positive asymmetric treatment eventually leads workers to form accurate expectations about wage offers. Immediately following the shock, average wages adjust by more than one point (from 6.13 to 7.36, column 1) while wage expectations remain relatively constant. Thus, average wage expectations are significantly below average wages (p-value $= 0.042$). However, experiencing higher wages for several periods causes expectations to adjust upward so that by round 15 average wages (7.27, column 1) and wage expectations (7.04, column 2) are no longer significantly different (p-value $= 0.701$). This demonstrates the ability of expectations to adjust via experience alone following economic shocks despite workers' complete lack of information regarding the shock.

We do not observe this same adjustment following wage surprises in negative shock sessions. Average wages (4.03, column 4) and wage expectations (5.4, column 5) are significantly different following a negative shock (p-value < 0.001). Despite workers and firms interacting for several post-shock periods together, and workers consistently receiving wages below their expectations, we find that significant differences between wages and wage expectations persist through round 15 (p-value $<$ 0.01). Importantly, average wages in this treatment are stable across post-shock rounds. This means that the difference between wages and wage expectations is driven by the failure of expectations to adjust rather than by an adjustment of wages.

Overall, this constitutes strong evidence that expectations adjustment is not symmetric in the presence of information frictions. People are willing to adjust expectations upward to meet unexpected wage surprises but expectations are more strongly anchored and sluggish to adjust whenever wage surprises are negative. This asymmetric adjustment aligns with Bewley (1999, 2007), who show via survey evidence that a primary component of downward nominal wage rigidity is an expressed concern from managers that lowering wages will destroy worker morale and reduce effort.

Result 8: *Announcing endowment shocks instantly shifts workers' wage expectations while information frictions lead to a sluggish and asymmetric adjustment; wage expectations and actual wages eventually converge following positive shocks but do not following negative shocks. Instead, the wage expectations of workers remain significantly higher than the actual wage.*

Table 3: Average wage offers and wage expectations [in elicitation rounds]

	Positive Asymmetric			**Negative Asymmetric**		
	(1) Wage Offer	(2) Expected Wage	(1) - (2) Difference p-value	(4) Wage Offer	(5) Expected Wage	(4) - (5) Difference p-value
Round 5	5.81	5.54	0.610	5.68	5.87	0.612
Round 10	6.13	5.59	0.248	5.68	6.18	0.283
Round 11	7.36	6.04	0.042	4.03	5.4	0.000
Round 15	7.27	7.04	0.701	3.84	5	0.000
Pairs	22	22	-	32	32	-
	Positive Full Info.			**Negative Full Info.**		
Round 5	6.64	6.71	0.787	6.78	6.42	0.402
Round 10	6.67	6.75	0.871	6.84	7.21	0.360
Round 11	8.57	8.5	0.911	4.39	4.81	0.080
Round 15	9.21	8.78	0.312	4.78	4.72	0.839
Pairs	28	28	-	33	33	-

Notes: Wage offer is the wage offered by the firm in the elicitation rounds. Expected wage is the wage that workers expect to receive in each elicitation rounds. We exclude pairs where the firm did not pass the asymmetric test. We report the p-values of a paired t-test for the equality of means.

Similarly, our evidence is consistent with the idea of wage ratcheting observed in Kaur (2019), where positive shocks generate persistently higher nominal wages but negative shocks do not lead to nominal wage cuts. In her setting, individuals believe nominal wage cuts are unfair and lead to effort reductions. Our evidence on expectation adjustment shows that wage hikes lead to quick upward updating while the opposite is not true for wage cuts under information frictions. Arguably, this rapid updating might anchor workers expectations and reduce tolerance for wage cuts in the future.

Table 4: Effort responses and wage expectations in round 11

	(1) Effort Choice (pos. shock)	(2) Effort Choice (pos. asym. shock)	(3) Effort Choice (neg. shock)	(4) Effort Choice (neg. asym. shock)
Mean of dep. var.	1.440 ***	1.402 ***	1.330 ***	1.274 ***
	(0.242)	(0.326)	(0.120)	(0.179)
Post	0.559 ***	-0.737 ***	-0.253 **	-0.099
	(0.093)	(0.263)	(0.127)	(0.115)
Post×Above	0.023	1.137 ***	0.138	0.082
	(0.186)	(0.325)	(0.209)	(0.447)
Post×Below	-1.266 ***	0.319	-0.176	-0.446***
	(0.178)	(0.330)	(0.223)	(0.146)
Male	0.222	0.133	0.183	0.292
	(0.283)	(0.465)	(0.201)	(0.224)
Nash eq.	0.536 ***	-0.678	-0.876***	-0.726***
	(0.152)	(0.449)	(0.324)	(0.248)
Observations	308	242	363	352

Notes: Results from random-effects linear regressions that include controls indicating gender and if a worker ever played Nash equilibrium in an elicitation period. *Above* indicates wage offers above, *Below* wage offers below and *Equal* corresponds to pairs with offers equal to workers' expectation in round 11. Standard errors clustered at the individual level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We next explore how the misalignment of wages and expectations impact effort choices. To do this, we first classify firm-worker pairs into three groups (Above, Below, or Equal) according to their wage offers relative to wage expectations in round 11. Using these classifications, we look for heterogeneous effects in effort choices using the same specification as in our previous analysis. We report results from this exercise in Table 4 where our main coefficients of interest are Post×Above and Post×Below, which report how effort responds to the surprise component of wages that are above or below the worker's wage expectation. We first consider results from our positive shock treatments. Whenever workers do not expect wage hikes following a positive shock, they strongly reciprocate wage increases (1.137 units, column 2). This effect is not present under symmetric information where workers expect wage increases following positive shocks. Instead, we see that workers strongly punish wages that fall short of their new, higher wage expectations whenever they know of the positive endowment shock (-1.266 units, column 1). This suggests that a rationale for firms to increase wages under symmetric information is avoiding retaliatory effort cuts when expectations are not met, rather than to elicit a higher effort via reciprocal motives. In contrast, when information frictions mask firm's intentions, workers who receive a wage above expectations elicit a higher effort response.

In the negative shock treatments, we find that workers strongly punish wage cuts if they are unexpected and lead to shortfalls relative to expectations (-0.446, column 4). However, this same effect is not present in the negative, symmetric information treatment (-0.176, column 3). This suggests that workers are more lenient with wage short falls whenever they expect (and can rationalize) wage cuts. It's worth remembering that the average wage is indeed quite similar in the post-shock rounds of both negative shock treatments. We can see that the large decrease in effort following the endowment shock in our negative asymmetric information treatment (-0.48, Figure E7b) is driven largely by workers retaliating against unexpected wage cuts (-0.44).

Taken together, our analysis in this section suggests that information frictions are more costly in terms of efficiency following negative shocks because sluggish adjustment of expectations leads to prolonged bouts of punitively low effort.

4 Estimating the Model and Evaluation

In simple structural estimation exercise, we quantify the out-of-sample predictive power of our model (equations (1) and (2)) relative to a baseline model that incorporates a static reference wage and does not respond to economic shocks:

$$e^*(w) = \alpha \left(\frac{w-a}{b}\right)^{\beta} \tag{5}$$

where a and b are constants to keep the benchmark comparable to our model. We first use data from both pre-shock elicitation rounds to structurally estimate the parameters of both models. Next, we predict out-of-sample post-shock effort levels and wages using our previous estimates. Finally, we compare these predictions to actual wage and effort choices and compute a mean squared predicted error (MSPE) for each subject. Although our model has an additional degree of freedom due to the dynamic reference wage, it is worth mentioning that prediction improvements are not mechanical, as we impose a functional form on the reference wage.[22] If our functional form is misspecified, that would cause lower predictive power for our suggested model.

We perform this estimation exercise using only subjects from symmetric information treatments. This is because the two models make almost identical predictions under asymmetric information - a prediction fully supported by our data - since the endowment remains unchanged from the worker's perspective following the endowment shock.[23] For details on the structural estimation identification and procedure, refer to Appendix section C.

4.1 Post-Shock Effort Predictions

We summarize results of effort predictions in Figure 4, which displays cumulative distribution functions (CDFs) of individual-level MSPEs for workers (a and b) and firms (c and d) using the benchmark model (red solid line) and our model (blue solid line). The corresponding vertical lines denote the medians of the two distributions.[24]

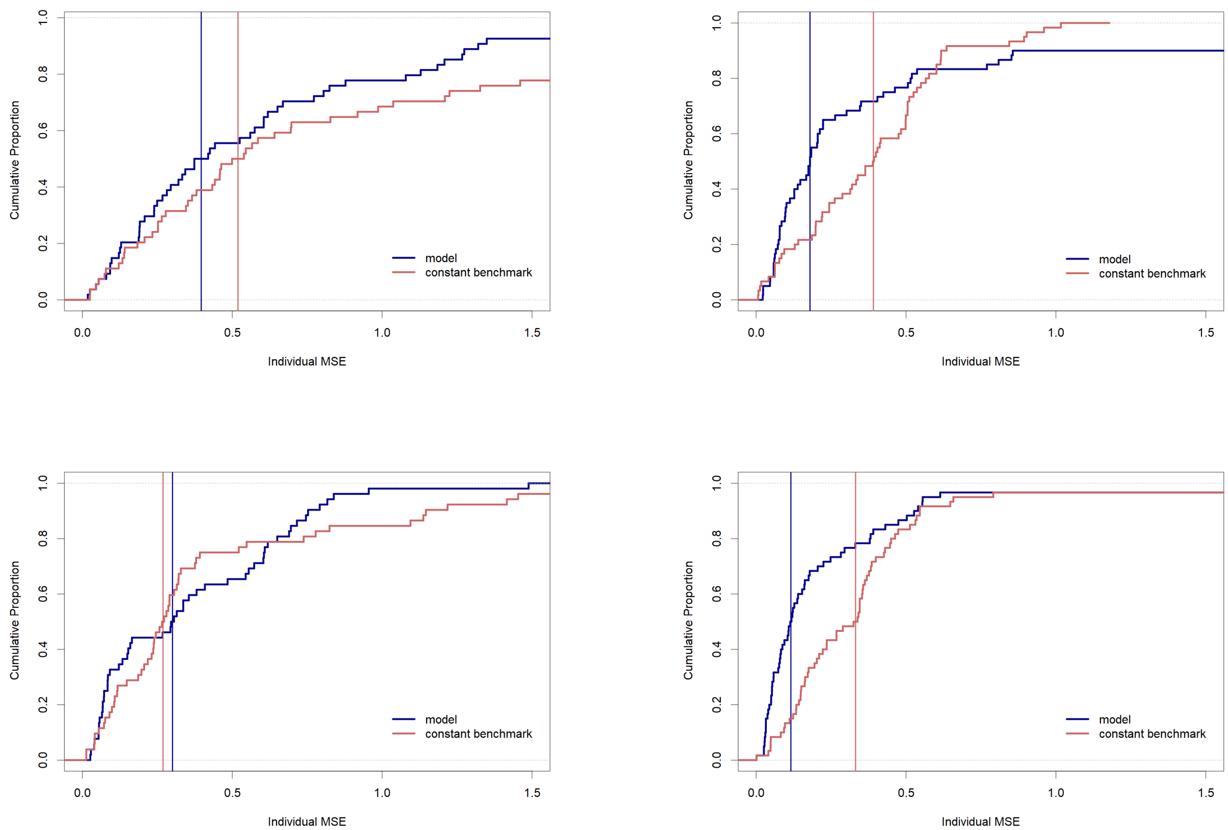
Figure 4 provides compelling evidence that our model outperforms the benchmark model in predicting both firms' beliefs about worker effort and workers' actual effort response functions. Our model reduces the median individual MSPE of effort response predictions by 24% following a positive endowment shock (4a) and by 54% following a negative shock (4b) relative to the benchmark model (one-sided Wilcox test p-values are 0.078 and 0.012 respectively). Likewise, our model reduces the median individual MSPE of predictions about firm beliefs by 65% after a negative shock (4d) relative to the benchmark model (one-sided Wilcox test p-values is <0.001). However, we find that both models predict firm beliefs equally well following a positive endowment shock (4c, one-sided Wilcox test p-value is 0.409).

[22] In particular, we assume the reference wage is a constant fraction of the firm's endowment.

[23] The only differences arise from two pairs in the positive asymmetric condition that chose to reveal the shock to their counterpart by offering a wage larger than 12.

[24] We evaluate medians rather than means to ensure that a few extreme outliers are not driving our results.

Figure 4: Individual Mean Squared Prediction Error (MPSE) for Effort Choices



Notes This figure shows CDFs of mean squared prediction errors (MSPEs) in rounds 11 and 15 for workers (panels a and b) and firms (panels c and d). Red dashed CDFs correspond to the benchmark model. Solid blue CDFs correspond to our relative wage model. Red vertical lines shows the median of the MSPEs from the benchmark model and the blue vertical line the median of the MSPEs from the relative wage model.

4.1.1 Post-Shock Wage Prediction

Next, we use our estimates for the firms in the pre-shock periods to obtain a prediction for wage offers in the post-shock periods and compare this to observed wage offers. Table 5 shows that our model does remarkably well in predicting shifts in wage offers – there is no significant difference between observed wages (column 1) and wages predicted by our model (column 2) in any of our four treatments.

Figure 4 shows that accounting for changes in economic conditions matters for two reasons. First, our model allows reference wages to adjust when economic conditions change whereas the benchmark model does not. Intuitively, we account for the idea that workers will share shortfalls and expect to share windfalls whenever they are exogenous. Second, effort accounts for wages relative to the firm's endowment. This captures the idea that a person's degree of altruism can depend on the perceived wealth of a firm and hence, why information about that wealth matters.

Table 5: Predicted wage comparison [in post-shock rounds]

	(1) **Wage offer**	(2) **Model prediction**	(1) - (2) **Model deviation**	(4) **Benchmark prediction**	(1) - (4) **Benchmark deviation**
Positive	9.06	9.08	0.980	8.47	0.215
Positive asym.	7.86	7.41	0.544	-	-
Negative	4.74	4.78	0.909	5.45	0.063
Negative asym.	4.06	4.26	0.495	-	-

Notes: This table compares predicted wages for our model (2) and a benchmark model (4) to actual wage offers (1) in all the post-shock rounds but the last period. Columns (3) and (5) report p-values of the difference between wage offers and model predictions respectively.

We find that firms correctly predict both of these channels, which is why our model better predicts firm's beliefs about effort response functions. Because firms act on these beliefs, our model also outperforms the benchmark model in predicting wages. More subtly, what we learn from Figure 4 is that information structure helps determine *how* changes in economic conditions impact wage-effort dynamics.

Finally, we want to emphasize that there is no tradeoff for using our model instead of the benchmark model. Out-of-sample predictions from our model are always at least as good as those from the benchmark model – usually, they are much better. More generally, it may make sense to adapt behavioral models of the labor market to account for changes in economic conditions and relative information sets.

5 Conclusion

This paper explores the role of information frictions in an experiment intended to capture key aspects of a labor market setting. We first build intuition using a toy model about how information frictions can significantly impact wage-effort dynamics. Our key insight is that workers evaluate wages relative to a reference wage, which itself depends on economic conditions. Because of this relationship, information frictions regarding economic conditions play a significant role in determining how wage-effort dynamics evolve in response to economic shocks. We then incorporate permanent endowment shocks, information frictions, and belief elicitation into a gift-exchange game to demonstrate these insights empirically.

Our results suggest workers evaluate wages according to a *state-contingent* reference wage. These state-contingent reference wages adjust instantaneously in response to news about endowment shocks so that subsequent wage changes are not a surprise and lead to little, if any, changes in effort. However, this dynamic reverses whenever endowment shocks and wage changes surprise uninformed workers. When surprises are positive, the uninformed worker's effort responds about twice as strongly as the informed workers. If the surprise is negative, the uninformed worker cuts effort four times more than the informed worker.

Experience alone can remove the wedge between wage expectations and wages. However, adjustment depends critically on the direction of wage surprises. Expectations fully adjust by our second post-shock elicitation in round 15 following positive wage surprises. In contrast, workers are more reluctant to adjust their reference wage in response to negative wage surprises. Because of this, wages and wage expectations remain significantly different in our second post-shock elicitation round. This is true even though, in both instances, average wages remain relatively stable across post-shock elicitation rounds. This asymmetric adjustment helps rationalize both wage ratcheting and downward nominal wage rigidity as empirical regularities.

Accounting for economic shocks and information frictions almost always improves the out-of-sample predictive power of the analogous behavioral labor market model. An implication is that models of reference dependence should allow for information frictions, especially when concerned with the dynamics of reference dependence.

Our results imply that differences in institutional features that create firm-level heterogeneity in information frictions may themselves lead to differences in wage frictions. For example, wage dynamics might differ between public and private firms, since the latter retain discretion and the former are compelled to reveal information. These features may also be cultural and lead to peer-to-peer information gaps. For example, recent work documents the reluctance to discuss wages between coworkers (Cullen and Perez-Truglia, 2018). Our results on the asymmetric adjustment of reference wages could rationalize this reluctance since only low-paid workers will likely revise their reference wages.[25] This aligns with evidence on pay secrecy and effort provision from Nosenzo (2013) and Cullen and Perez-Truglia (2022). Our results are also consistent with Kline et al. (2019), who show that workers capture about 30 percent of patent-induced productivity shocks but that this share is roughly double for workers who were with a firm when they originally filed for the patent and that wages for new entrants do not respond to these shocks. To some extent, employees familiar with the patent filing and decision, or who participated in its development, have more information about the potential impact of the patent decision on firm profitability.

Though our results shed light on the allocation of surplus resulting from exogenous economic shocks, it is often true that real-world surplus is endogenous. This can obviously influence both the workers information set and the firm's outlook on welfare distribution. A meaningful extension then of our work might be to endogenize economic volatility while revealing different amounts of information to workers about output and how output relates to their own effort provision. Importantly, the asymmetric adjustment of wage expectations that we observe highlights the importance of the economic context underpinning a firm's wage decisions.

[25] Charness and Kuhn (2007) provide evidence that revealing wages may not lead to lower effort.

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Appendix (for online publication)

A Additional Tables and Figures

Table A6: Wage offers pre/post shock

	(1) Wage offer (pos. shock)	(2) Wage offer (asym.pos. shock)	(3) Wage offer (neg. shock)	(4) Wage offer (asym.neg. shock)
Mean of dep.var	6.874	6.932	6.670	6.015
Post	2.258***	1.221**	-2.100***	-1.729***
	(0.470)	(0.512)	(0.210)	(0.174)
Final Round	0.631	-0.394	-0.091	-0.708**
	(0.574)	(0.743)	(0.293)	(0.298)
Male	-0.948**	-0.258	0.553*	-0.525
	(0.481)	(0.421)	(0.310)	(0.488)
Nash Eq.	2.572***	-2.307***	-1.172	-0.351
	(0.492)	(0.581)	(1.017)	(0.496)
Trial Round Avg.	0.256	0.463***	-0.140	0.206
	(0.176)	(0.147)	(0.103)	(0.191)
Observations	560	440	660	640

Notes: Results from a random-effects linear regression. We control for the final round, gender, if the pair played the Nash equilibrium strategy (defined as the worker selecting effort equal to 0 for all possible wages in at least one elicitation) and for the corresponding average wage offered during the trial rounds. We exclude pairs where the firm did not pass the asymmetric test. Clustered standard errors at the individual level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Effort Choices pre/post shock

	(1) Effort Choice (pos. shock)	(2) Effort Choice (asym.pos. shock)	(3) Effort Choice (neg. shock)	(4) Effort Choice (asym.neg. shock)
Mean of dep.var	1.179**	0.809	1.440***	1.617***
Post	0.135	0.116	-0.125	-0.481***
	(0.112)	(0.115)	(0.088)	(0.099)
Final Round	-0.240	-0.049	-0.238*	-0.275***
	(0.214)	(0.141)	(0.126)	(0.097)
Male	0.127	-0.418	0.343**	0.086
	(0.181)	(0.424)	(0.156)	(0.208)
Nash Eq.	0.335**	-0.976**	-0.687**	-0.610***
	(0.157)	(0.461)	(0.311)	(0.213)
Trial Round Avg.	0.050	0.150	-0.030	-0.046
	(0.089)	(0.097)	(0.057)	(0.068)
Observations	560	440	660	640

Notes: Results from a random-effects linear regression. We control for the final round, gender, if the pair played the Nash equilibrium strategy (defined as the worker selecting effort equal to 0 for all possible wages in at least one elicitation) and for the corresponding average wage offered during the trial rounds. We exclude pairs where the firm did not pass the asymmetric test. Clustered standard errors at the individual level in parentheses. * $p < 0.10$, *** $p < 0.05$, *** $p < 0.01$.

Table A8: Beliefs and effort response changes in post-shock periods (symmetric information)

	(1)	(2)	(3)	(4)
	Beliefs diff.	Beliefs diff.	Effort Resp. diff.	Effort Resp. diff.
	(pos. shock)	(neg. shock)	(pos. shock)	(neg. shock)
Mean of dep. var.	0.215 *	-0.100	0.180	-0.155 *
Wage	-0.087 ***	0.106 ***	-0.066***	0.118***
	(0.013)	(0.021)	(0.016)	(0.019)
Observations	228	264	228	264

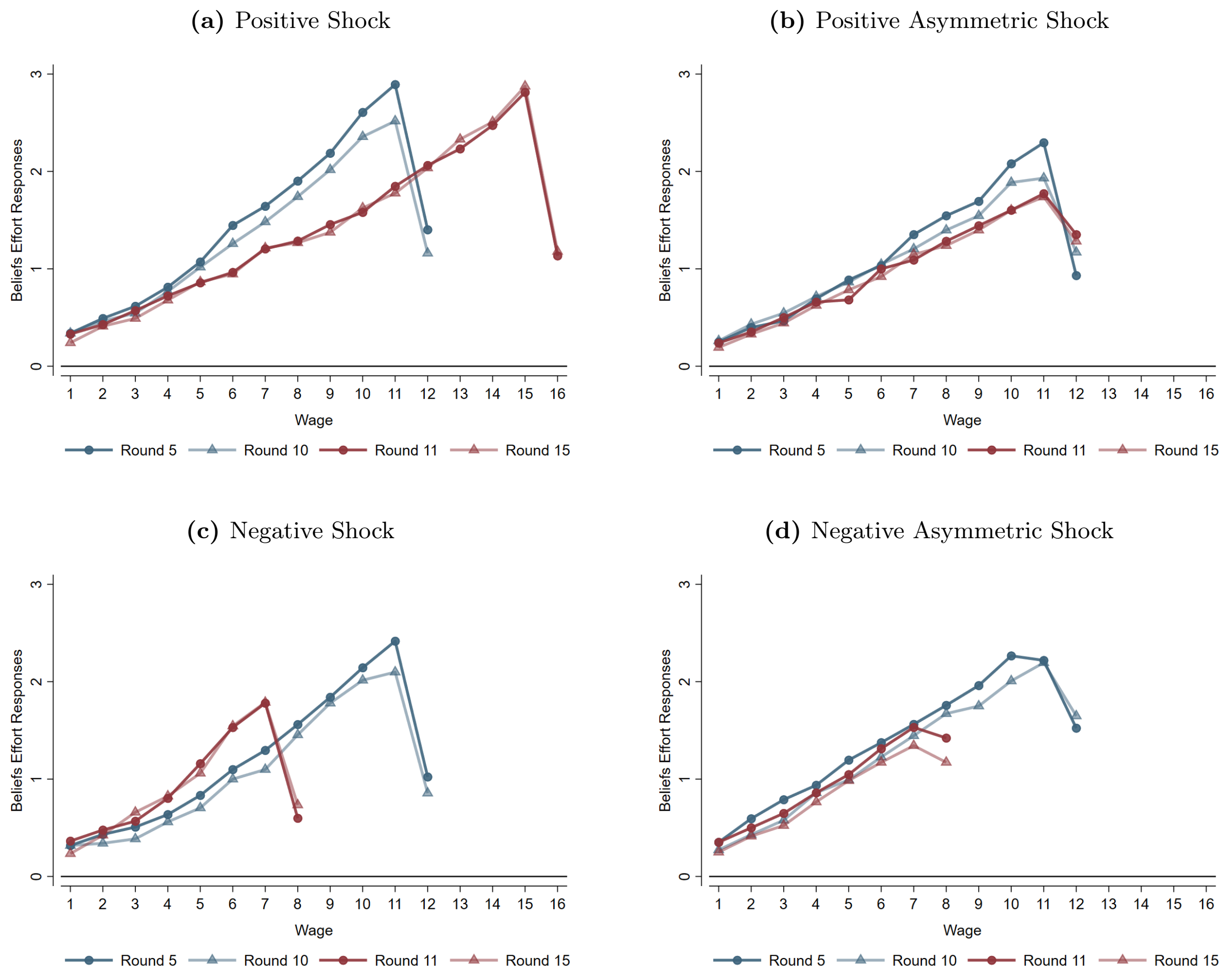
Notes: Results from a random-effects linear regression. Rounds 5 and 10 are pooled together, similarly with rounds 11 and 15. We control for the maximum wage (the one that exhausts the endowment), gender, if the pair played the Nash equilibrium strategy (defined as the worker selecting effort equal to 0 for all possible wages in at least one elicitation), and for the corresponding average wage offered during the trial rounds. Clustered standard errors at the individual level in parentheses. * $p < 0.10$, *** $p < 0.05$, *** $p < 0.01$.

Table A9: Beliefs and effort response changes in post-shock periods (asymmetric information)

	(1)	(2)	(3)	(4)
	Beliefs diff.	Beliefs diff.	Effort Resp. diff.	Effort Resp. diff.
	(asym.pos.shock)	(asym.neg.shock)	(asym.pos.shock)	(asym.neg.shock)
Mean of dep.var.	0.038	0.082	0.044	0.028
Wage	-0.016*	-0.003	-0.007	-0.009
	(0.009)	(0.014)	(0.010)	(0.006)
Observations	216	288	216	432

Notes: Results from a random-effects linear regression. Rounds 5 and 10 are pooled together, similarly with rounds 11 and 15. We control for the maximum wage (the one that exhausts the endowment), gender, if the pair played the Nash equilibrium strategy (defined as the worker selecting effort equal to 0 for all possible wages in at least one elicitation), and for the corresponding average wage offered during the trial rounds.. We exclude pairs where the firm did not pass the asymmetric test. Clustered standard errors at the individual level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

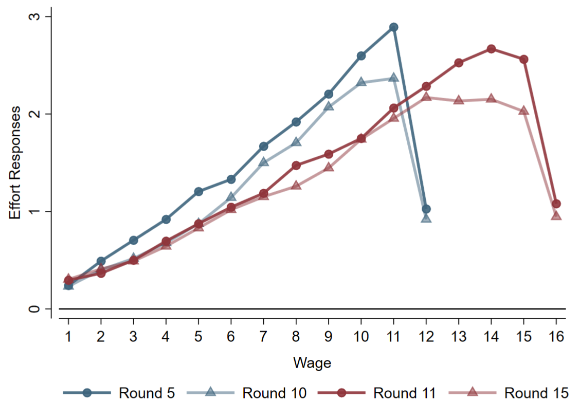
Figure A5: Firms' beliefs about effort responses by elicitation round



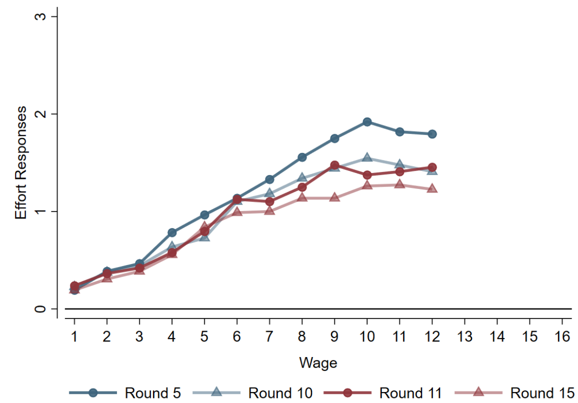
Notes: This figure shows the average belief about effort responses for each possible wage. Drops corresponds to wage offers that exhaust the firm's endowment. We exclude two firms that chose to reveal the shock in the positive asymmetric condition.

Figure A6: Workers' effort responses by elicitation round

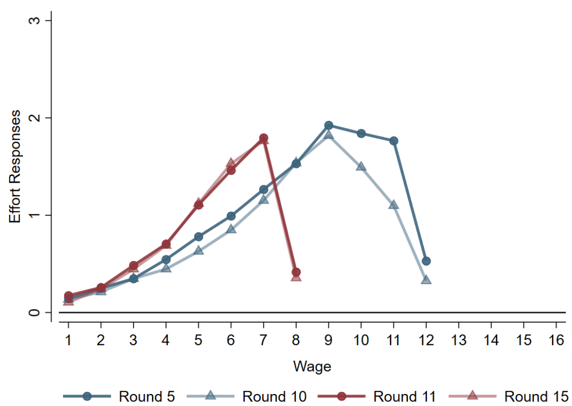
(a) Positive Shock



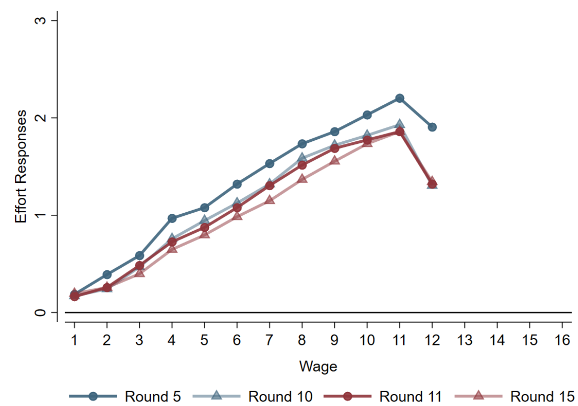
(b) Positive Asymmetric Shock



(c) Negative Shock



(d) Negative Asymmetric Shock



Notes: This figure shows the average effort response for each possible wage. Drops corresponds to wage offers that exhaust the firm's endowment. We exclude two workers for which the firm chose to reveal the shock in the positive asymmetric condition.

Table A10: Firms Payoffs Pre/Post Shock

	(1) Firm payoff (pos. shock)	(2) Firm payoff (asym.pos. shock)	(3) Firm payoff (neg. shock)	(4) Firm payoff (asym.neg. shock)
Mean of dep.var	6.145	8.918	5.566	6.890
Post	2.633***	4.169***	-2.471***	-4.269***
	(0.448)	(0.911)	(0.324)	(0.508)
Final Round	-2.356**	0.784	-0.899***	-0.663**
	(1.098)	(1.063)	(0.308)	(0.335)
Male	2.019**	-2.342	0.838**	1.241
	(0.860)	(3.027)	(0.428)	(0.868)
Nash Eq.	-1.214	-4.965	-2.722***	-2.921***
	(0.861)	(3.239)	(0.884)	(1.025)
Trial Round Avg.	-0.162	0.197	0.010	-0.444*
	(0.414)	(0.522)	(0.167)	(0.238)
Observations	560	440	660	640

Notes: Results from a random-effects linear regression. We control for the final round, gender, if the pair played the Nash equilibrium strategy (defined as the worker selecting effort equal to 0 for all possible wages in at least one elicitation) and for the corresponding average wage offered during the trial rounds. We exclude pairs where the firm did not pass the asymmetric test. Clustered standard errors at the individual level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Workers Payoffs Pre/Post Shock

	(1) Worker payoff (pos. shock)	(2) Worker payoff (asym.pos. shock)	(3) Worker payoff (neg. shock)	(4) Worker payoff (asym.neg. shock)
Mean of dep.var	4.304***	3.034***	6.296***	3.174***
Post	1.948***	0.970**	-1.870***	-1.081***
	(0.339)	(0.440)	(0.140)	(0.142)
Final Round	0.886*	-0.287	0.070	-0.510*
	(0.489)	(0.721)	(0.251)	(0.281)
Male	-1.223***	0.644	0.029	-0.693*
	(0.450)	(0.685)	(0.231)	(0.415)
Nash Eq.	2.285***	-0.850	-0.578	0.322
	(0.466)	(0.708)	(0.744)	(0.433)
Trial Round Avg.	0.178	0.191*	-0.118	0.286*
	(0.153)	(0.114)	(0.086)	(0.150)
Observations	560	440	660	640

Notes: Results from a random-effects linear regression. We control for the final round, gender, if the pair played the Nash equilibrium strategy (defined as the worker selecting effort equal to 0 for all possible wages in at least one elicitation) and for the corresponding average wage offered during the trial rounds. We exclude pairs where the firm did not pass the asymmetric test. Clustered standard errors at the individual level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Firms profits and wage expectations

	(1) Firm Payoff (pos. shock)	(2) Firm Payoff (pos. asym. shock)	(3) Firm Payoff (neg. shock)	(4) Firm Payoff (neg. asym. shock)
Mean of dep. var.	6.801**	8.512*	5.475***	9.338***
Post	3.369***	6.003***	-2.576***	-4.339***
	(0.831)	(2.255)	(0.501)	(1.229)
Post×Above	0.054	-1.113	0.216	-0.565
	(0.948)	(2.435)	(0.716)	(1.624)
Post×Below	-1.936	-6.193**	0.076	0.172
	(1.187)	(3.054)	(0.591)	(1.328)
Final Round	-2.356**	0.784	-0.899***	-0.663**
	(1.100)	(1.065)	(0.309)	(0.336)
Male	2.195***	-2.563	0.848*	1.177
	(0.806)	(2.727)	(0.440)	(0.884)
Nash eq.	-1.760**	-4.881*	-2.734***	-2.867**
	(0.773)	(2.923)	(0.898)	(1.139)
Trial Round Avg.	-0.123	0.046	0.015	-0.446*
	(0.383)	(0.476)	(0.174)	(0.246)
Observations	560	440	660	640

Notes: Results from a random-effects linear regression. We control for the final round, gender, if the pair played the Nash equilibrium strategy (defined as the worker selecting effort equal to 0 for all possible wages in at least one elicitation) and for the corresponding average wage offered during the trial rounds. We excluded pairs where the firm did not pass the asymmetric test. Type *Above* is a dummy that corresponds to pairs with wage offers above workers' expectation in round 11. Type *Below* to those with wage offers below and *Equal* corresponds to pairs with offers equal to expectations. Clustered standard errors at the individual level in parentheses. * $p < 0.10$, *** $p < 0.05$, *** $p < 0.01$.

	Type Classification [as a percentage of pairs]						
	Round 11			Round 15			
	Above	**Below**	**Equal**	**Above**	**Below**	**Equal**	**#Pairs**
Positive	46.4	39.3	14.3	32.1	17.9	50.0	28
Positive asymmetric	63.6	18.2	18.2	45.5	27.3	27.3	22
Negative	30.3	51.5	18.2	24.2	30.3	45.5	33
Negative Asymmetric	9.4	71.9	18.8	12.0	62.5	25.0	32

Notes: This table shows the percentage of pairs in each treatment where the firm offers a wage above, below, or equal to the worker's expected wage in rounds 11 and 15.

B Details for Theoretical Section

In this section, we provide additional details on the derivation of optimal wage and effort functions with both symmetric and asymmetric information. Additionally, we consider hypotheses under two alternative specifications for the firm's profit function.

B.1 Solving our stylized model

We solve the model via best response analysis. Under the assumption that $c(e)$ is strictly convex, $U(\cdot)$ is strictly concave in e and the worker can always find a utility-maximizing level of effort e for any offered wage. Assuming the cost function $c(e) = 0.5e^2$, the worker's optimal effort response to a wage w and some reference wage $w_0(R)$ is:

$$e_i^*\left(w, R, w_0(R)\right) = \alpha_j \left(\frac{\mu + w - w_0(R)}{R}\right)^{\beta_j} \tag{6}$$

which is increasing in wages, decreasing in the reference wage, and decreasing in beliefs about the firm's endowment.[26] As mentioned previously, μ represents the normalized utility value when the worker receives the reference wage.

Substituting $e_i^*(w, R)$ into the firm's profit function $(R-w)e$ and solving for the optimal wage yields the firm's best-response function:[27]

[26]We assume $\mu \geq w_0(R) - \underline{w}$ to ensure non-negative effort responses.

[27]Our experimental results suggest that firms do form accurate beliefs about the effort response function, understand how a change in the endowment reshapes the effort response function, and form accurate beliefs about the worker's expected wage.

$$w^*(e^*) = \frac{\beta_j R + w_0(\tilde{R}) - \mu}{1 + \beta_j} \quad (7)$$

Notice first that Equation 4 is decreasing in μ. This is because, all else equal, higher values of μ yield a larger effort response for any wage w. Second, Equation 4 is increasing in the firm's endowment R and the worker's belief about the endowment $\tilde{R}$ through $w_0(\cdot)$. Intuitively, this says wages will always adjust in the direction of the endowment shock. Firms share positive shocks in order to elicit a reciprocal effort response from workers and share negative shocks expecting only a proportional decrease in effort. Further, firms understand that $w_0(R)$ is order preserving – the worker will expect a wage increase following a positive shock ($w_0(R)$ shifts up) but will tolerate a wage cut following a negative shock ($w_0(R)$ shifts down). These two channels highlight the relevance of information – or absence thereof – regarding economic conditions in determining optimal wages.

B.2 Alternative profit functions

Firms in our experiment face an induced utility function of the form $\pi = (R - w)e$. Though this functional form was and remains common in relevant pieces of literature, researchers have begun to consider alternative functional forms to address the concern that our chosen functional form implies the marginal product of effort is declining in wages.

If one re-frames our chosen functional form in terms of a Cobb-Douglas function where effort and the firm's endowment are complementary then the idea that the marginal product of effort declines in wages is not problematic. This is because allocating capital stock to the worker as payment before production reduces the firm's capital stock that interacts with worker effort.

There are at least two other functional forms suggested to us by reviewers, and conference and seminar participants. First, we could have considered a functional form where only the wage is multiplied by effort so that $\pi = R - we$. Second, we could have considered a functional form where effort scales only the endowment so that $\pi = Re - w$. The former addresses any concerns about effort multiplying the endowment while the second addresses the concern that the marginal product of the effort is declining in the wage.

Considering $\pi = R - we$

Suppose the firm's profit function is $\pi = R - we$ and the firm's goal is to maximize profits with respect to wage. Clearly, the optimal wage in this setting is $w = 0$, which yields $\pi = R$. This is because effort no longer scales endowment and so the firm has no incentive to induce positive effort. In fact, this functional form implies that firm profits are declining in effort. Further, subjects in our experiment could coordinate on $w = R,\ e = 0$ so that the firm earns $\pi = R - R * 0 = R$ and the worker earns $U(w, e) = w - c(e) = R - 0 = R$. Intuitively, this means the worker and firm could magically double and split the endowment without any labor input. For these two reasons, we chose not to adopt this functional form for firm profits.

Considering $\pi = Re - w$

Suppose instead that the firm's profit function is $\pi = Re - w$ and again the firm wants to maximize profits with respect to wage. If the firm accounts for the worker's optimal effort response $e_i^*(w, \tilde{R}) = \alpha_j \left(\frac{\mu + w - w_0(\tilde{R})}{\tilde{R}} \right)^{\beta_j}$, then the profit-maximizing wage is

$$w^*(w, R, w_o(R)) = (\frac{1}{\alpha\beta})^{\frac{1}{\beta-1}} R + w_o(R) - \mu$$

.

The concern is whether behavior ought to be qualitatively different when we assume alternative specifications for firm profits. This is equivalent to asking whether our hypotheses would change qualitatively if we assume $\pi = Re - w$ (rather than $\pi = (R - w)e$). Since hypotheses 2 and 4 concern worker effort, we can ignore them. Thus, we focus here on hypotheses 1 and 3.

We first consider Hypothesis 1, which says that wage changes are positively correlated with changes in the firm's endowment when information is symmetric. Since $\alpha,\ \beta$ are both strictly positive, we have that changes in w^* and R are positively correlated. Thus, hypothesis 1 is qualitatively equivalent under this alternative specification.

Next, we consider Hypothesis 3, which says that information frictions moderate wage changes following an endowment shock. Assume a firm's endowment changes from $R, R',\ R' > R$. The difference in the corresponding wage change for fully-informed workers is given by

$$
\begin{aligned}
w^*(w, R', w_o(R')) - w^*(w, R, w_o(R)) &= (\frac{1}{\alpha\beta})^{\frac{1}{\beta-1}} R' + w_o(R') - \mu - \Big[(\frac{1}{\alpha\beta})^{\frac{1}{\beta-1}} R + w_o(R) - \mu\Big] \\
&= (\frac{1}{\alpha\beta})^{\frac{1}{\beta-1}} (R' - R) + (w_o(R') - w_o(R))
\end{aligned}
$$

With asymmetric information, we would have

$$
\begin{aligned}
w^*(w, R', w_o(R)) - w^*(w, R, w_o(R)) &= (\frac{1}{\alpha\beta})^{\frac{1}{\beta-1}} R' + w_o(R') - \mu - \Big[(\frac{1}{\alpha\beta})^{\frac{1}{\beta-1}} R + w_o(R) - \mu\Big] \\
&= (\frac{1}{\alpha\beta})^{\frac{1}{\beta-1}} (R' - R) + w(_o(R) - w_o(R)) \\
&= (\frac{1}{\alpha\beta})^{\frac{1}{\beta-1}} (R' - R)
\end{aligned}
$$

Since we assume that $w_o(R)$ is strictly increasing in R, we have that the wage increase following a positive endowment shock is strictly smaller with asymmetric information by the amount $-(w_o(R') - w_o(R)) < 0$. By similar logic, we would predict that wages fall by less following a negative endowment shock if workers are uninformed. For example, if the endowment changes from R to R', $R' < R$ then $-(w_o(R') - w_o(R)) > 0$.

C Structural Estimation

This section provides details on our structural estimation.

C.1 Identification and Estimation

We first focus on the worker's problem. Log-linearization of equation (3) yields the following equation:

$$
ln(e^*(w, R)) = ln(\alpha) + \beta \; ln \left(\frac{\mu + w - w_0(\tilde{R})}{\tilde{R}} \right) \tag{8}
$$

Because we directly observe w and $\tilde{R}$, to be able to estimate α, β in equation (8) via ordinary least squares (OLS) we need measures of both μ and $w_0(\tilde{R})$. Since neither of those is directly observable, we first assume $\mu = 0$. This assumption biases our estimates of α (i.e. the intercept) but it does not affect β (i.e. the slope) since μ is a constant that shifts the effort response function but does not impact its curvature. Biasing α does not reduce the goodness of fit of either model and we can still estimate wage offers since the optimal wage does not depend on α. This leaves only the issue that we do not directly observe $w_0(\tilde{R})$.

One possibly solution is to proxy $w_0(\tilde{R})$ using wage expectations gathered from workers during elicitation rounds. However, this proxy leads to many instances in our data where $w - w_0(\tilde{R}) < 0$ for which the log of the ratio is undefined in $\mathbb{R}$. To circumvent this issue, we instead proxy $w_0(\tilde{R})$ using the wage $\overline{w}_0(\tilde{R})$ such that $e^*(w, R) = 0$ whenever $w \leq \overline{w}_0(\tilde{R})$. For $\overline{w}_0(\tilde{R})$ to be a valid proxy, it should react to endowment shocks in a way consistent with our theoretical assumptions about $w_0(\tilde{R})$. We provide evidence of this in table C13, which shows that our proxy measure reacts to endowment shocks in the same direction as fair wages under symmetric information and does not change under asymmetric information.

Table C13: Reference wage proxy [in elicitation rounds]

	(1) **Pre-shock proxy**	(2) **Post-shock proxy**	(2)-(1) **Diff.**
Positive	1.24	2.13	0.88
Positive asymmetric	1.38	1.47	0.08
Negative	1.50	0.92	-0.58
Negative Asymmetric	1.04	1.21	0.16

Notes: This table shows the average wage for which workers choose a zero effort level in the elicitation rounds. $\overline{w}_0(R)$ is defined as the wage such that $e^*(w, R) = 0$ whenever $w \leq \overline{w}_0(R)$. Rounds 5 and rounds 10 are pooled together. Similarly for rounds 11 and 15. As in the remaining of our structural exercise, we exclude pairs with an all-zero effort elicitation.

While we obtain $\overline{w}_0(\tilde{R})$ directly from pre-shock data, we need to be able to estimate this value following an endowment shock. This means we must assume a functional form for $\overline{w}_0(\tilde{R})$ in order to produce post-shock estimates of wages, beliefs, and effort that do not rely on post-shock data. Thus, we further assume $\overline{w}_0(\tilde{R}) = \gamma\tilde{R}$, which says that our proxy of the reference wage is simply a constant fraction of the endowment. Note that if this is a misspecification of $\overline{w}_0(\tilde{R})$ then it will penalize our model while leaving unaffected the benchmark model.

In our benchmark model the worker's effort response function does not react to endowment shocks or incorporate information frictions. We model this by replacing $w_0(\tilde{R})$ and $\tilde{R}$ with two constants. For simplicity, we let these two constants match the pre-shock values of $\overline{w}_0(\tilde{R})$ and $\tilde{R}$, respectively.[28]

$$ln(e^*(w,R)) = ln(\alpha) + \beta\ ln\left(\frac{w-a}{b}\right) \tag{9}$$

Intuitively, this benchmark model implies that neither changes in the endowment nor information regarding those changes impact a worker's effort response function. The models are otherwise identical.

We start by estimating individual-level values of α and β using pre-shock elicitation data for workers (from effort response function) and firms (from belief elicitation) separately.[29] We then predict out-of-sample post-shock values for wages, beliefs, and effort with both models using these parameter estimates. Comparing the resulting distributions of individual-level MSPEs determines which model provides better out-of-sample predictions.

C.1.1 Post-Shock Wage Prediction

Next, we use our estimates for the firms in the pre-shock periods to obtain a prediction for wage offers in the post-shock periods and compare this to observed wage offers. Our proxy $\overline{w}_0(\tilde{R})$ almost certainly introduces downward bias in our wage predictions since the levels are likely lower than actual reference wages. However, we can recover firm's actual beliefs of worker's reference wage $\hat{w}_0(\tilde{R})$ if we assume that the firm set wages optimally given their beliefs (this is supported by evidence in section 3.4). Given this assumption, we can then use our estimates of β and pre-shock wage offers to recover the firm's beliefs from equation (1) and its benchmark analog.[30] Finally, we assume the same functional form as before $\hat{w}_0(\tilde{R}) = \gamma\tilde{R}$ so we can do out-of-sample predictions for

[28]Though the values of a, b do not influence the out-of-sample predictive power of this benchmark model, this approach to selecting their values ensures that pre-shock values of α, β are consistent across the two models that we consider.

[29]In many instances, all effort responses or beliefs are 0 in a given elicitation. We drop these observations from the estimation since logs are undefined. In addition, we do not take into account the responses for when the wage offer exhausts the endowment for the same reason. For a discussion about how to deal with logs and zeros in regression models see Young and Young (1975) and Bellego et al. (2021).

[30]The benchmark equation only differs in that $w_0(\cdot)$ is substituted by a constant.

changes to the endowment.

D Instructions

This subsection of the Appendix provides the instructions used in our experiment. There was one minor difference in the instructions used for symmetric and asymmetric information sessions. This difference - where we change what information workers receive at the end of a Decision Period - is highlighted in red text. This difference, which corresponds to a difference in end-of-period information for workers across information conditions, was necessary to create our desired information frictions. Importantly, our baseline periods of play across information conditions confirm that this difference did not meaningfully affect workers' effort decisions.

We provided instructions in two phases. First, we provided subjects with a one-page document meant to outline the basics of the game they would play, how their decisions would affect their earnings, and how we would construct their final payoffs. We followed this with a basic comprehension quiz, delivered on-screen. Subjects could not proceed without answering all questions in the comprehension quiz correctly. Next, we provided subjects with more detailed information the the two types of periods they would experience. We again followed this set of instructions with an on-screen comprehension quiz.

D.1 Instructions - Basic Information Sheet

Basic Information

Formation of Pairs:

We will randomly assign you to one of two roles for this experiment: employer or worker. You will be informed about your role at the beginning of this experiment. Once we assign you to one of these two roles we will then randomly form employer-worker pairs. Your role (employer or worker) will not change during this experiment. Additionally, the composition of worker-employer pairs will not change during the paid portion of this experiment. That means that you will always make decisions with the same worker/employer during the paid portion of this experiment

Anonymity:

This is an anonymous game. No other participant will know your identity during this experiment. Your counterpart will know your role (whether you are an employer or worker) and will see your decisions. However, your counterpart cannot link these decisions to you personally. No other participant (other than your counterpart) will know your role or be able to view your decisions.

Experiment Length:

- This experiment lasts for 20 paid periods. A period ends after both players (worker and employer) make all relevant choices.
- However, you will play 3 additional trial periods to get used to the game. Trial periods are not paid. It is important that you take these seriously so that you understand the game. This will help you maximize your earnings during paid periods. After the 3 trial periods, the 20 paid periods will begin automatically

Description of the 20 periods

There are two types of periods during the paid portion of this experiment:

- *Decision Periods:* workers and employers make some decisions.
- *Questionnaire Periods:* first, players will respond to some questions; then, they will proceed to making their decisions. You will start the experiment by playing a Decision Period and you will be notified when you are playing a Questionnaire Period.

Payment for this experiment:

- We will pay you in cash at the end of this experiment.
- The computer will randomly select one Decision Period and one Questionnaire Period out of the 20 periods. We will pay you for these two randomly selected periods. This is in addition to your $10 show-up fee.
- Thus, your final payment will include your earnings for the two randomly selected periods and the show-up fee.

D.2 Instructions - Description of Periods

1.Description of a Decision Period

Steps

Each *Decision Period* proceeds as follows:

1. The employer receives an endowment and then sets a wage for the worker
2. The worker learns the wage offered and then selects a level of effort in response
3. Both subjects learn their payoff for that period
4. The game proceeds to either another Decision Period or to a Questionnaire Period

Roles and Actions

- **Employer:**
  - A subject assigned to the role of employer' initially receives an endowment of \$12. The employer will use the per-period endowment to pay a wage to the worker for that period.
  - The wage must be some number between \$1 and the total endowment in increments of \$1. Thus, an employer could pay a wage of \$1,\$2,\$3, , etc.
  - Whatever portion of the endowment the employer keeps is multiplied by the effort provided by the worker and then paid to the employer.
  - Formally, the employer will earn:

$$\text{Employer's payoff} = (\text{Endowment} - \text{Wage}) * \text{Effort}$$

  - **An Example:** Suppose that the employer decides to pay the worker a wage of \$4. Further, suppose that the worker selects effort of 1.5. With these hypothetical values, the employer earns $(12 - 4) * 1.5 = 8 * 1.5 = \$12$.
  - Note that only what is left of the initial endowment is multiplied by the level of effort chosen by the worker. Thus, paying the whole endowment as a wage would yield a payoff of zero for the employer no matter the level of effort the worker selects in response.
  - At the end of each period, the employer will learn both the worker's effort level and his/her own payoff.

- The employer cannot accumulate money throughout periods. This means the employer can use only his/her endowment to set a wage in each period.

- **Worker:**
  - A subject assigned to the role of 'worker' will receive a wage offer in each period from the employer.
  - In each period, after receiving the employer's wage offer, the worker will select a level of costly effort for that period. The higher the effort, the higher is the cost of effort.
  - The level of effort can be: [0, 0.25, 0.5, 1, 1.25, 1.50,..., max effort]. Note that a worker cannot select a level of effort that will cause a negative payoff.
  - Workers have the following payoff function:

$$\text{Worker's payoff} = \text{Wage} - \text{Cost of Effort}$$

    The cost of effort is computed according to the following function and displayed before confirming the effort choice:

$$\text{Cost of effort} = .5 * e^2$$

  - Suppose an employer offers a wage of \$6 and the worker exerts an effort level of 2 in response.Then the worker earns $6 - (0.5 * 2^2) = 6 - 2 = \$4$
  - At the end of each period, the worker will learn his/her own payoff. Note: For symmetric information treatments, subjects also learned the employers payoff, which we informed them of here.
  - The worker cannot accumulate money throughout periods. That is, each period the worker will receive a wage offer and go through the steps described above.

2. Description of a Questionnaire Period

Steps

Each *Questionnaire Period* proceeds as follows:

1. Both the worker and the employer respond to some questions
2. Both the worker and the employer make their Decisions

3. Payoffs of the worker and the employer for that period are displayed
4. The game proceeds to either another Questionnaire Period or to a Decision Period.

Further details on points 1 and 2 will be given during the game and displayed on screens.Instructions are role specific. We display these instructions during each questionnaire period. However, the instructions for a questionnaire period are the same for all questionnaire periods.

The software will notify you when you are playing a Questionnaire Period. You will start the game by playing a Decision Period.

3. Final remarks:

If you have any further questions, please raise your hand and an experimenter will visit your station to answer your question privately.

4. Summary of Basic Instructions:

- We will randomly assign you to one of two roles: employer or worker. Your role will not change during the experiment.
- We randomly assign one worker to one employer. This is the employer-worker pair. These pairs do not change during the experiment.
- The game lasts for 20 periods (plus 3 trial periods) and they are distinguished into Decision Periods and Questionnaire Periods.
- In Each Decision period, the employer receives an endowment and uses this endowment to pay a wage to the worker.
- Once the worker receives a wage offer, the worker will respond by selecting a level of costly effort.
- Employer earnings for each period are (Endowment - Wage)*Effort. -Worker earnings for each period are (Wage - Cost of Effort).
- The employer receives a new endowment in each period, and the employer and the worker cannot accumulate money across periods.
- The employer selects a new wage each Decision Period. The worker selects a new level of

costly effort each Decision Period.

- During Questionnaire Periods, both players respond to some questions displayed on their respective screens. The software will always notify you when you are playing a Questionnaire Period.
- 1 Decision Period and 1 Questionnaire Period will be randomly chosen and the payoffs in these periods, plus your show-up fee, form you final payment.

E Additional Robustness Treatment & Sample Justification

E.1 Robustness Treatment

We run an additional treatment using a sample of 26 subjects from the online platform Prolific. Seven pairs were assigned to the negative shock condition, whilst the remaining six were assigned to the positive shock condition. All pairs face the asymmetric information condition, yet in these additional sessions we explicitly stated in the instructions that the firm's endowment may or may not change. In particular, we added the following bullet point when describing roles and payoffs in the instructions:

The endowment of the employee might change at some point during the study without the worker knowing

In addition, we included a control check at the end of the experiment to determine whether participants suspected that the employers endowment had changed or not. Specifically, we asked at the end of the study:

Over the course of the experiment, did you think that the endowment of the employer you were paired with had changed?

The potential answers were: no change, it increased, and it decreased.

Results from the control check question

Half of the participants (50%) reported that the employers initial endowment remained constant throughout the experiment, with this response evenly distributed across both negative and positive shock conditions. Another 38.89% stated that it increased, again split equally between conditions, meaning only half of the participants correctly identified the actual shock they experienced. The remaining 11.11% believed the endowment had decreased.

Overall, we report that 70% of the participants in the experiment was wrong at guessing the shock and its direction, while 30% correctly guessed them.

Changes relative to our laboratory study:

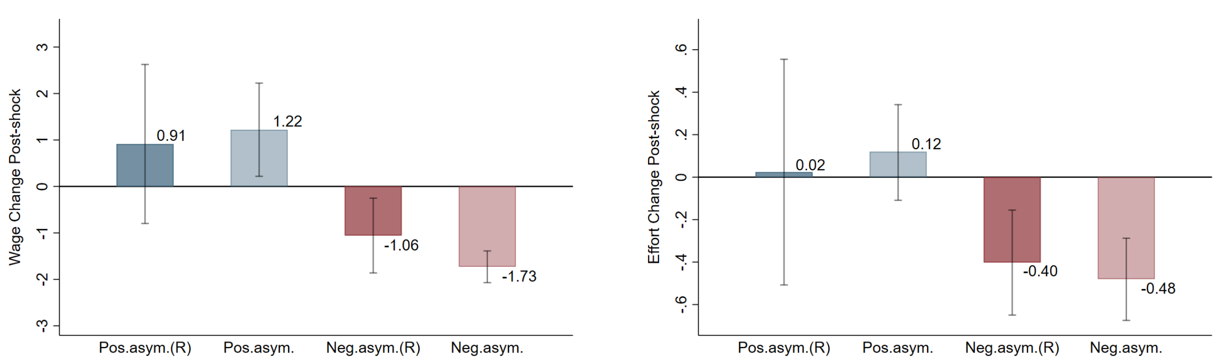
Since this additional robustness treatment was online, a couple of minor adjustments were made. Because participants are paired online, we set a time limit of 30 seconds to 2 minutes for each question to ensure active participation. After the time limit, if a question remains unanswered, pairs proceed to the next round. We drop all pairs where at least one participant skipped more than 1 question. This led to dropping one pair in the negative condition and two pairs in the positive condition. This attrition is orthogonal to the treatment assignment, as in all cases, the unanswered questions occurred prior to round 11, when the shock to the firm took place. Following this screening, we are left with a final sample of 13 pairs, which is roughly a quarter of our original sample size from the previous main treatments, 7 pairs assigned to the negative condition and 6 to the positive one. Among the remaining pairs able to miss at most one round, the share of questions missed is 4.93% in the first 10 rounds and 4.62% in the last 10 rounds, suggesting that the shock did not have an impact on this dimension.

Replication of our previous asymmetric information treatments:

We replicate in Figure E7 our main analysis already reported in the manuscript, but using only asymmetric information treatments from original and recent robustness treatments, denoted with (R).

Figure E7: Average Wage and Effort Responses by treatment (Assymetric Information)

(a) Wage Responses to Shocks　　**(b)** Effort Response to Shocks

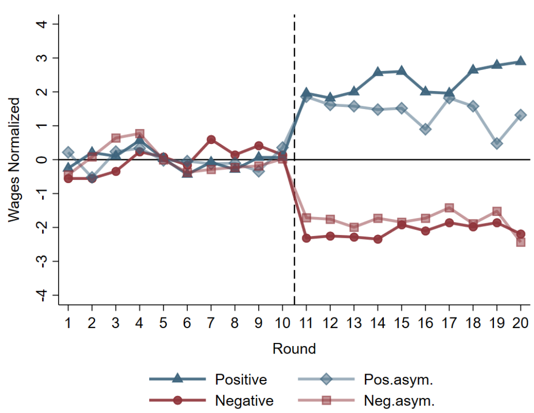


Notes: We depict the average change in wages (a) and effort (b) following endowment shocks in each of our four asymmetric information treatments (bars) surrounded by 95% confidence intervals estimated using a series of random-effects linear regressions that group pre- and post-shock data. We control for the final round, gender, if the pair played the Nash equilibrium strategy (defined as the worker selecting effort equal to 0 for all possible wages in at least one elicitation), and for the corresponding average wage offered during the trial rounds. Clustered standard errors at the individual level.

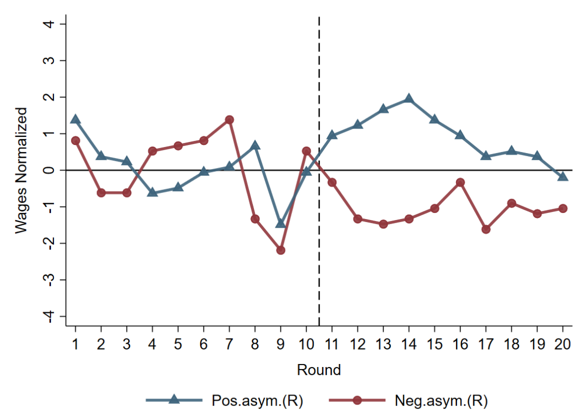
Overall, when looking at the change in wage levels, we find the point estimates to be quite similar, and in the negative case, still significantly different than zero despite the small sample. Hence, our main story remains: employers retain positive shocks and share negative ones, even when workers are aware of the possibility of a shock occurring. In terms of the effort responses, the similarity is even more striking. Small positive wage increases do not significantly alter the average effort responses, although the results are quite noisy. On the other hand, the small wage decrease leads to a punitive response from the workers, not only statistically significant but also comparable to our previous one. This is expected, as only a tiny fraction of workers believe that a negative shock has occurred, as indicated in our exit questionnaire.

Figure E8: Average Wage and Effort Responses by treatment

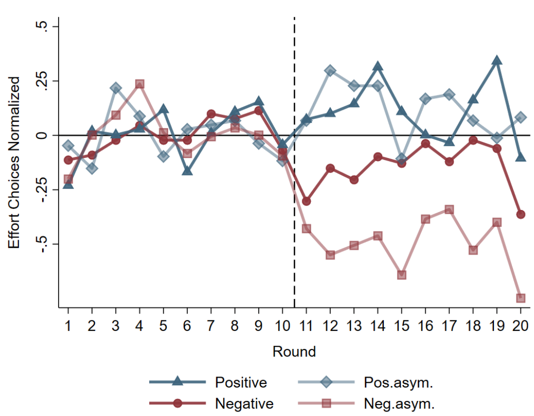
(a) Wage offers (Original Treatments)



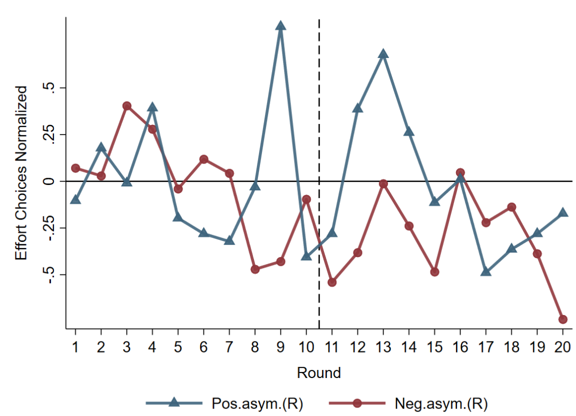
(b) Wage offers (Robustness Treatments)



(c) Effort responses (Original Treatments)



(d) Effort responses (Robustness Treatments)

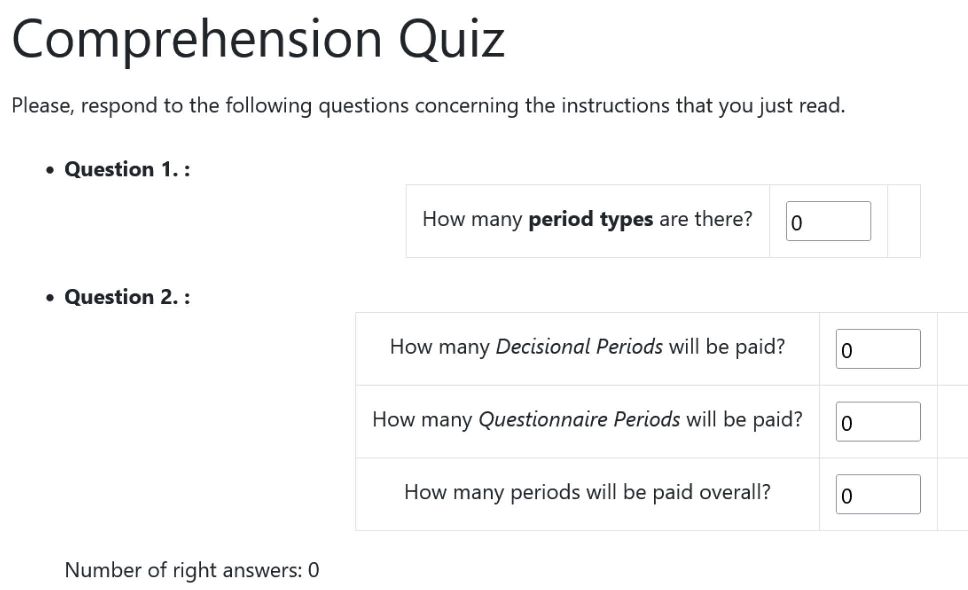


Notes: This figure shows the average wage (a,b) and effort choice (c,d) for each round. We normalize each variable by subtracting the pre-shock average and exclude pairs where the firm did not pass the asymmetric test.

E.2 Sample Justification

Our sample size was primarily determined by budget constraints at the time of data collection. To assess the sensitivity of our study design, we conducted a power analysis focusing on the main effect of the treatment condition. The calculation was based on 55 independent units, where each unit corresponds to the participants average outcome across repeated observations. This approach treats repeated measures at the participant level, rather than assuming independent observations, as done in the manuscript (see, for example, Figure 1). At 80% power and a 5% significance level, the minimum detectable effect size is Cohens $f^2 = 0.14$, which is considered a moderate effect size according to conventional benchmarks (Cohen 1988). This estimate provides a conservative reference point for evaluating the magnitude of the effects observed in our study.

F Screen Shots



Comprehension Quiz

Please, respond to the following questions concerning the instructions that you just read.

- **Question 1. :**

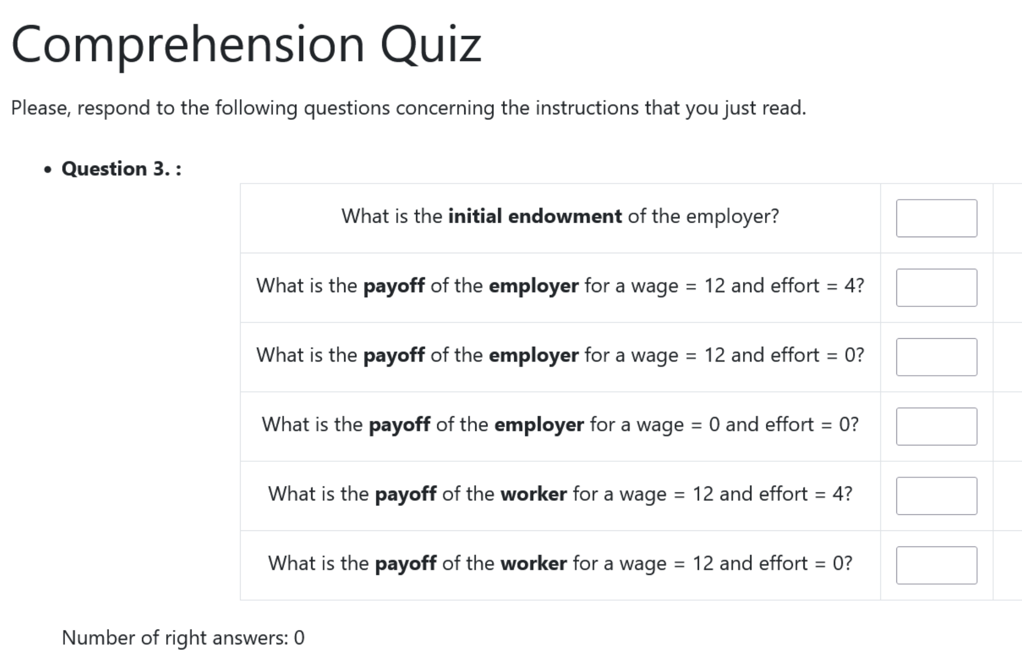
How many **period types** are there?	0

- **Question 2. :**

How many *Decisional Periods* will be paid?	0
How many *Questionnaire Periods* will be paid?	0
How many periods will be paid overall?	0

Number of right answers: 0

(e) First comprehension quiz



Comprehension Quiz

Please, respond to the following questions concerning the instructions that you just read.

- **Question 3. :**

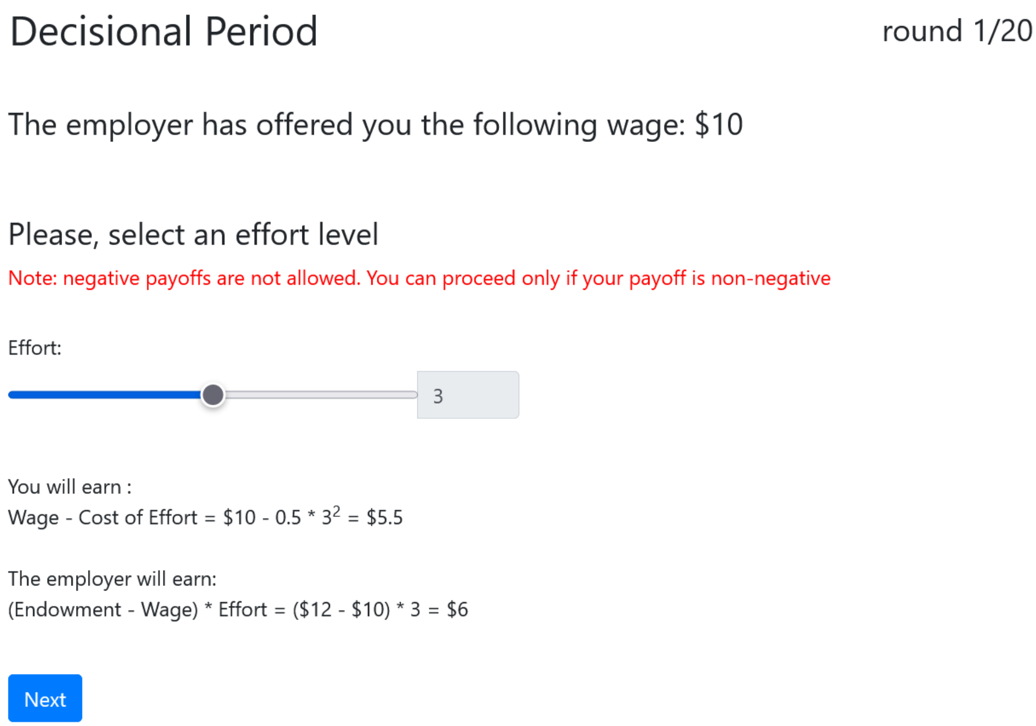
What is the **initial endowment** of the employer?	
What is the **payoff** of the **employer** for a wage = 12 and effort = 4?	
What is the **payoff** of the **employer** for a wage = 12 and effort = 0?	
What is the **payoff** of the **employer** for a wage = 0 and effort = 0?	
What is the **payoff** of the **worker** for a wage = 12 and effort = 4?	
What is the **payoff** of the **worker** for a wage = 12 and effort = 0?	

Number of right answers: 0

(f) Second comprehension quiz

Figure F9: Beliefs and Effort Responses to Shocks

Figure F10: Worker's effort selection screen



Notes: In asymmetric information treatments, the worker is not shown the firm's payoff.

Figure F11: Worker's effort selection screen

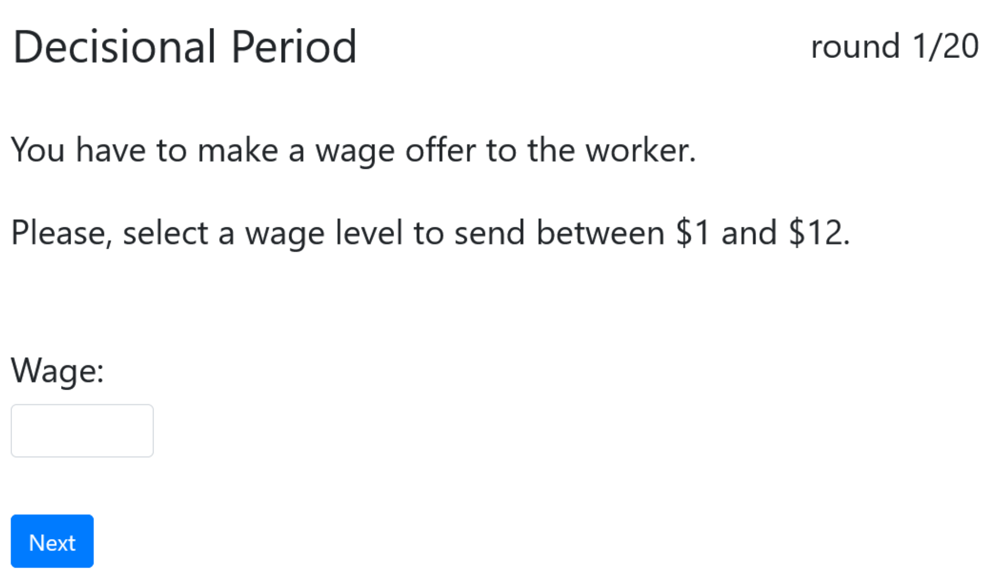
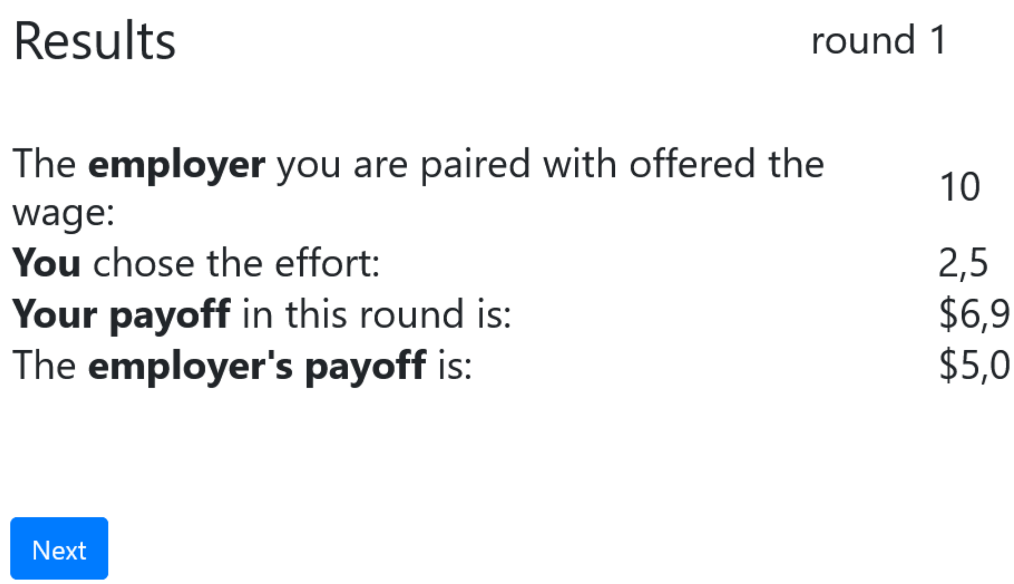
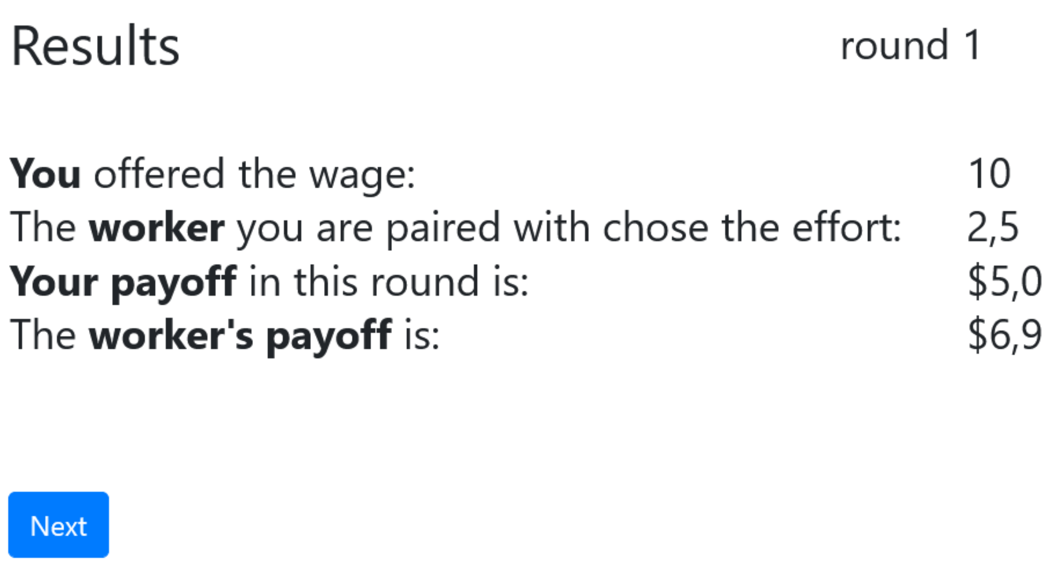


Figure F12: End-of-period information

(a) Firm's end-of-period information



(b) Worker's end-of-period information



Notes: In asymmetric information treatments, the worker is not shown the firm's payoff.

Change in Employer's Endowment

The employers' per-period endowment has changed from $12 to $8.

This change will last for the remainder of this experiment!

Please, note:

- The employer's payoff is now ($8 – Wage)* Effort.
- Therefore, the **highest** wage offer possible is now $8 rather than $12.
- This screen is being displayed **only** to **employers** in this experiment.
- Workers **do not learn** about this change.

Click on the button below to proceed.

Next

(c) Negative shock under asymmetric information

Change in Employer's Endowment

The employers per-period endowment has changed from $12 to $8.

This change will last for the remainder of this experiment!

Please, note:

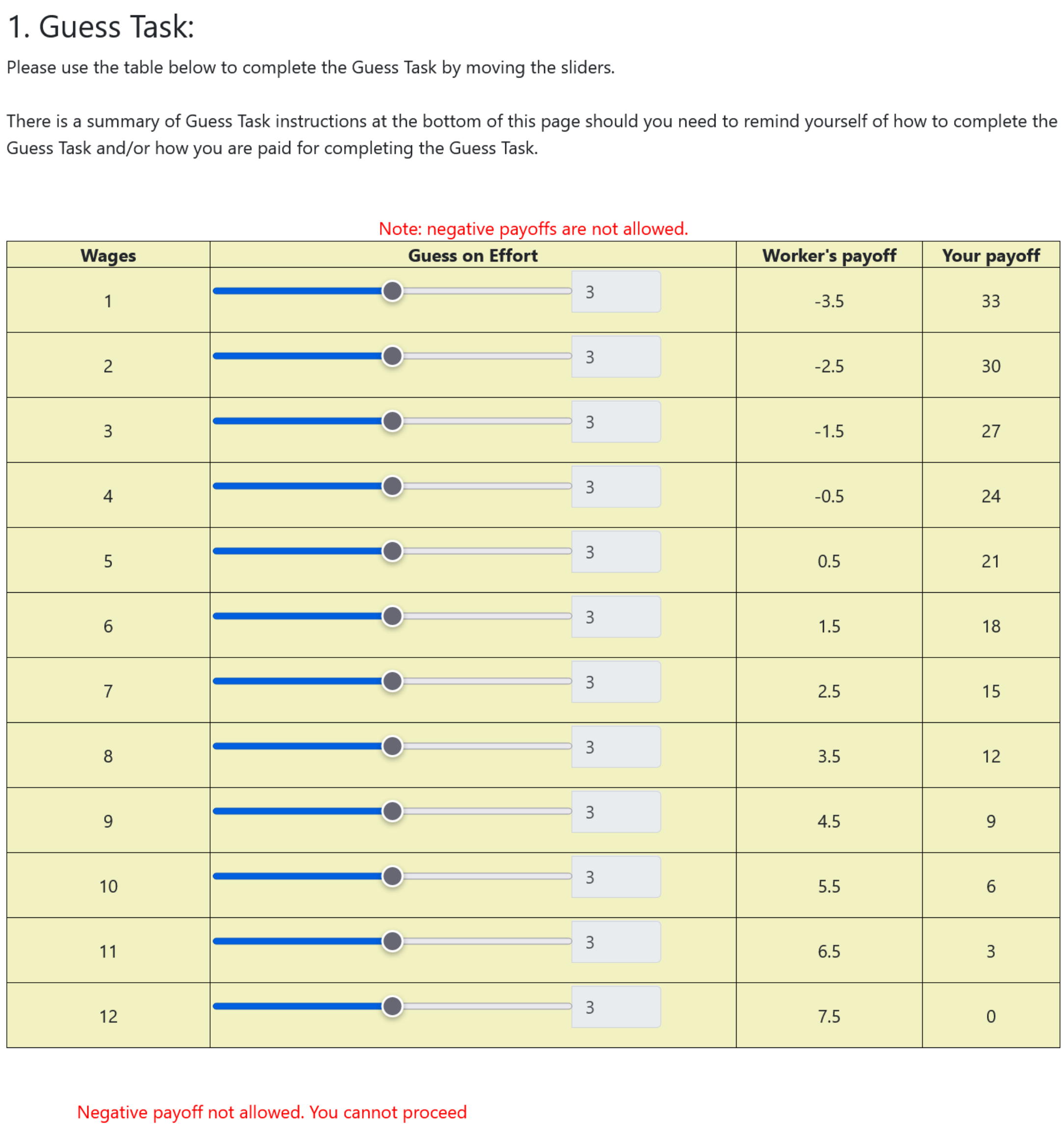
- The employer's payoff is now ($8 – Wage)* Effort.
- Therefore, the **highest** wage offer possible is now $8 rather than $12.
- This screen is being displayed to all employers and workers in this experiment.

Click on the button below to proceed.

Next

(d) Negative shock under symmetric information

Figure F13: Examples of endowment shock announcement



1. Guess Task:

Please use the table below to complete the Guess Task by moving the sliders.

There is a summary of Guess Task instructions at the bottom of this page should you need to remind yourself of how to complete the Guess Task and/or how you are paid for completing the Guess Task.

Note: negative payoffs are not allowed.

Wages	Guess on Effort	Worker's payoff	Your payoff
1	3	-3.5	33
2	3	-2.5	30
3	3	-1.5	27
4	3	-0.5	24
5	3	0.5	21
6	3	1.5	18
7	3	2.5	15
8	3	3.5	12
9	3	4.5	9
10	3	5.5	6
11	3	6.5	3
12	3	7.5	0

Negative payoff not allowed. You cannot proceed

Figure F14: Example of the belief elicitation screen seen by subjects sorted into the role of employer in periods 5, 10, 11 and 16

Figure F15: Example of the effort function elicitation seen by subjects sorted into the role of worker in periods 5, 10, 11 and 16

2. Strategy Task:

You will use the table below to provide your effort choice for each possible wage the employer could send you in this round. Note that in this task you will not known in advance the offered wage (as in previous Decisional Periods). That is:

- The employer will make a wage offer as usual, but this time it will not be displayed until the end of this task. You will learn the wage offer only after completing this task.
- You have to complete the table below with your effort response for each hypothetical wage level

How you earn money for the Effort Provision Task

- To compute your payoff for this task, the effort level corresponding to the wage offered by the employer will be chosen as effort response.

Example

- suppose you have completed the table below.
- Now, suppose that the employer offers a wage of $6 and that you agreed previously to provide effort 1.5 for a wage offer of $6 (that is the line 6 of table).
- Your payoff is computed as usual : Wage - Cost of Effort = ($6 - 0.5 * 1.5 ^{2}) = $4.9 in this example.

------------------------ Please, indicate your effort choices in the table below by moving the sliders.------------------

Note: negative payoffs are not allowed. You can proceed only if your payoff is non-negative

Hypothetical Wages	Your Effort	Your payoff	Employer's payoff
1	0.75	0.7	8.3
2	0.75	1.7	7.5
3	0.5	2.9	4.5
4	0.5	3.9	4
5	3	0.5	21
6	3	1.5	18
7	3	2.5	15
8	3	3.5	12
9	3	4.5	9
10	3	5.5	6
11	3	6.5	3
12	3	7.5	0

Next

Notes: In asymmetric information treatments, the column "Employer's payoff" is not shown.